

SSF Formulations

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In this script we demonstrate different approaches to including covariates in an SSF model.

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Load required packages

```
library(tidyverse)
packages <- c("amt", "sf", "terra", "RColorBrewer", "mgcv", "gratia")
walk(packages, require, character.only = T)
```

Import data and clean

```
buffalo_data <- read_csv("data/buffalo.csv")
```

New names:

Rows: 133161 Columns: 11

-- Column specification

```
----- Delimiter: "," chr
(2): node, dates dbl (7): ...1, lat, lon, height, accuracy, heading, speed dtm
(2): timestamp, DateTime
i Use `spec()` to retrieve the full column specification for this data. i
Specify the column types or set `show_col_types = FALSE` to quiet this message.
* `` -> `...1`
```

```
# remove individuals that have poor data quality or less than about 3 months of data.
# The "2014.GPS_COMPACT copy.csv" string is a duplicate of ID 2024, so we exclude it
buffalo_data <- buffalo_data %>% filter(!node %in% c("2014.GPS_COMPACT copy.csv",
                                                    2029, 2043, 2265, 2284, 2346))
```

```
buffalo_data <- buffalo_data %>%
  group_by(node) %>%
  arrange(DateTime, .by_group = T) %>%
  distinct(DateTime, .keep_all = T) %>%
  arrange(node) %>%
  mutate(ID = node)
```

```
buffalo_clean <- buffalo_data[, c(12, 2, 4, 3)]
colnames(buffalo_clean) <- c("id", "time", "lon", "lat")
attr(buffalo_clean$time, "tzone") <- "Australia/Queensland"
buffalo_clean$sex <- "f"
buffalo_clean$life_stage <- "adult"
head(buffalo_clean)
```

A tibble: 6 x 6

	id	time	lon	lat	sex	life_stage
	<chr>	<dtm>	<dbl>	<dbl>	<chr>	<chr>
1	2005	2018-07-25 10:04:02	134.	-12.3	f	adult
2	2005	2018-07-25 11:04:23	134.	-12.3	f	adult
3	2005	2018-07-25 12:04:39	134.	-12.3	f	adult
4	2005	2018-07-25 13:04:17	134.	-12.3	f	adult
5	2005	2018-07-25 14:04:39	134.	-12.3	f	adult
6	2005	2018-07-25 15:04:27	134.	-12.3	f	adult

```
tz(buffalo_clean$time)
```

```
[1] "Australia/Queensland"
```

```
buffalo_ids <- unique(buffalo_clean$id)

write_csv(buffalo_clean, "data/buffalo_clean.csv")
```

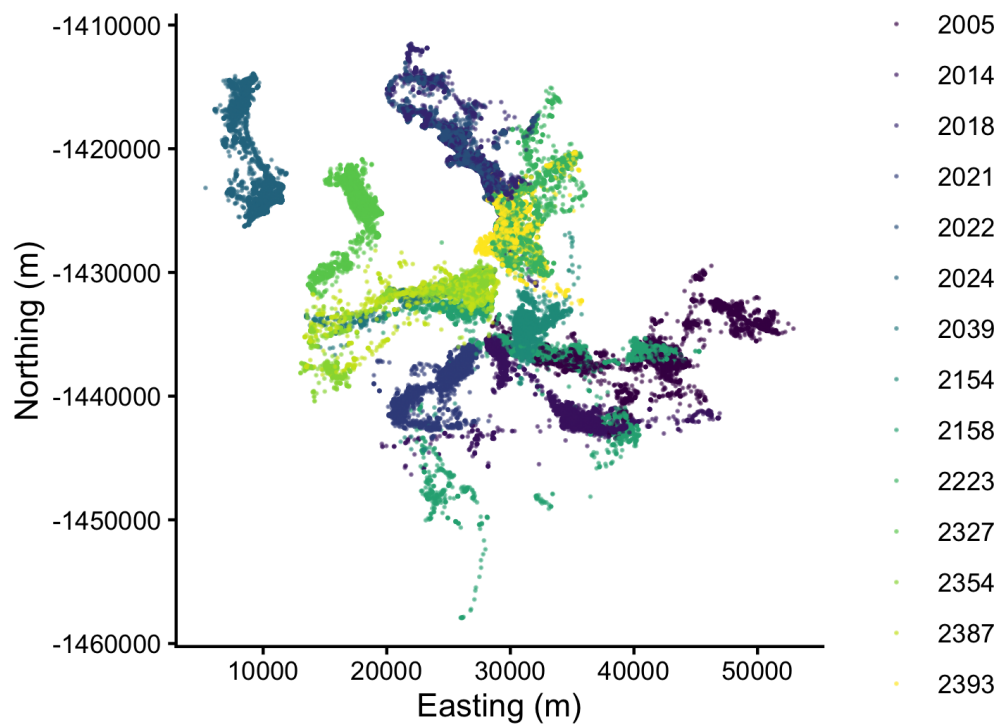
Create a step object

Use the `amt` package to create a trajectory object from the cleaned data.

```
buffalo_all <- buffalo_clean %>% mk_track(id = id,
                                           lon,
                                           lat,
                                           time,
                                           all_cols = T,
                                           crs = 4326) %>%
  transform_coords(crs_to = 3112, crs_from = 4326) # Transformation to GDA94 /
# Geoscience Australia Lambert (https://epsg.io/3112)
```

Plot the data spatially

```
buffalo_all %>%
  ggplot(aes(x = x_, y = y_, colour = id)) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  scale_colour_viridis_d() +
  theme_classic() +
  theme(legend.position = "right")
```



```
# ggsave("outputs/data_prep/buffalo_djelk_map.png",
#         width = 150, height = 150, units = "mm", dpi = 600)
```

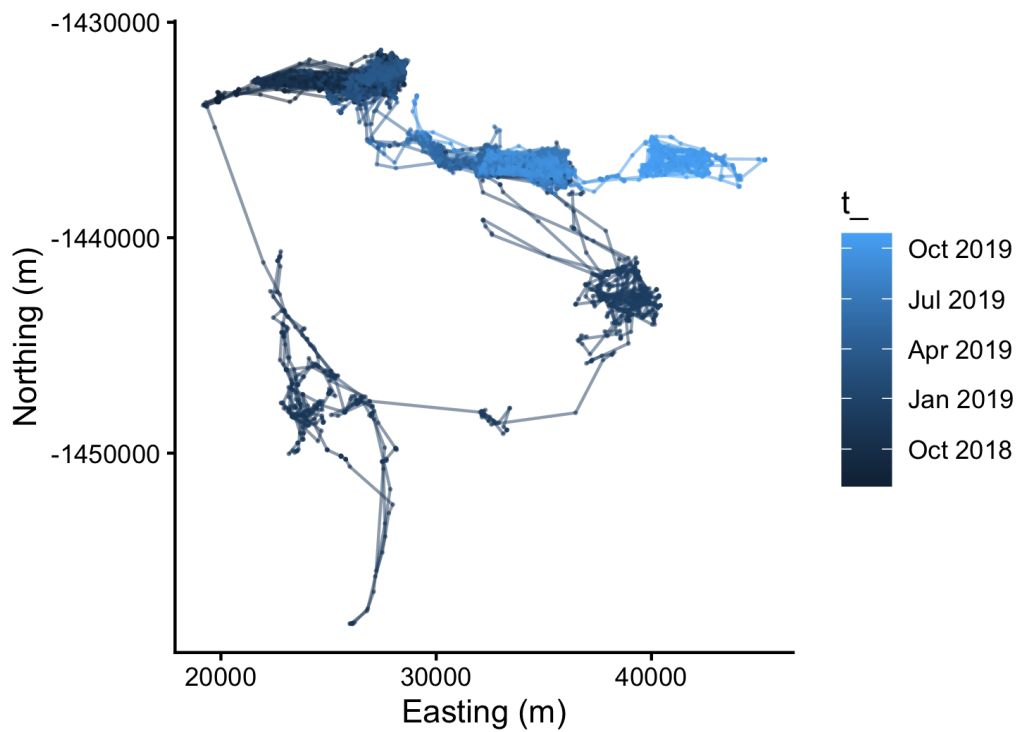
Pick out a single individual

```
which_buffalo <- "2158" # select a single buffalo ID

# create directory to save plots for each buffalo
plot_save_path <- paste0("outputs/buffalo_", which_buffalo)
dir.create(plot_save_path, showWarnings = FALSE)

buffalo_id <- buffalo_all %>% filter(id == which_buffalo)

buffalo_id %>%
  ggplot(aes(x = x_, y = y_, colour = t_)) +
  geom_path(alpha = 0.5) +
  geom_point(alpha = 0.5, size = 0.1) +
  coord_fixed() +
  scale_x_continuous("Easting (m)") +
  scale_y_continuous("Northing (m)") +
  theme_classic()
```

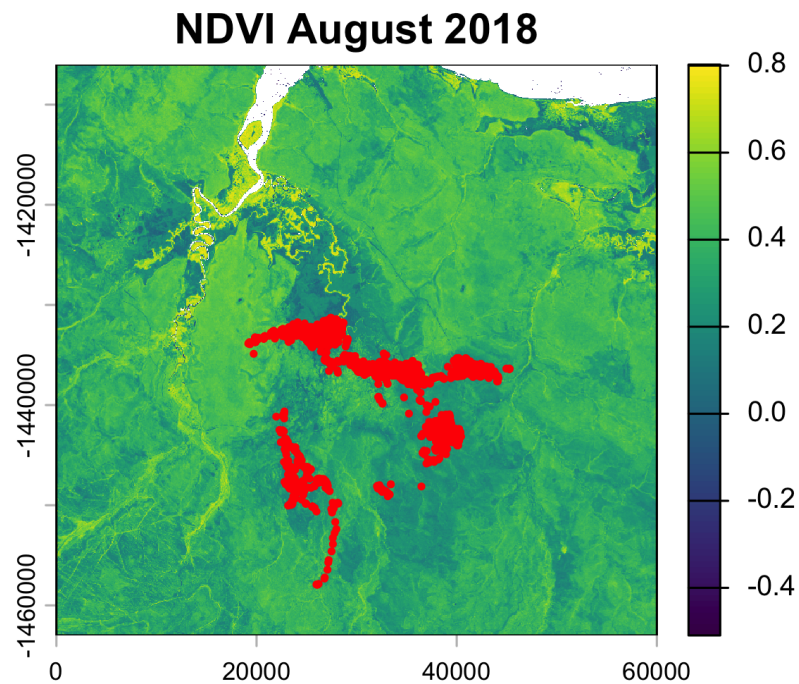


```
ggsave(paste0(plot_save_path, "/buffalo_id_", which_buffalo, "_map.png"),
       width = 150, height = 150, units = "mm", dpi = 600)
```

Import spatial covariate

Although NDVI changes over time, and we have access to monthly layers, we will just select a single month here.

```
ndvi <- rast("mapping/ndvi_aug_2018.tif")
plot(ndvi, main = "NDVI August 2018")
points(buffalo_id$x_, buffalo_id$y_, col = "red", pch = 16, cex = 0.5)
```



```
ndvi
```

```
class      : SpatRaster
size       : 2280, 2400, 1  (nrow, ncol, nlyr)
resolution : 25, 25  (x, y)
extent     : 0, 60000, -1463000, -1406000  (xmin, xmax, ymin, ymax)
coord. ref.: GDA94 / Geoscience Australia Lambert (EPSG:3112)
source     : ndvi_aug_2018.tif
name       : ndvi
min value  : -0.544105
max value  : 0.808655
```

Select local extent

We just want to look at an area centred on a single location, which might be combined with our movement probability surface to generate a next-step probability surface.

We'll centre the extent on an interesting looking part of the landscape, and then buffer it by 1500m (with a buffer of the cell resolution/2 to ensure there is a central cell), which contains most of the step lengths, and therefore most of the movement kernel.

```

cell_resolution <- 25 # in metres

buffalo_single_point <- buffalo_id[7500,] # select one of the locations

buffer <- 1500 + (cell_resolution/2) # buffer in metres

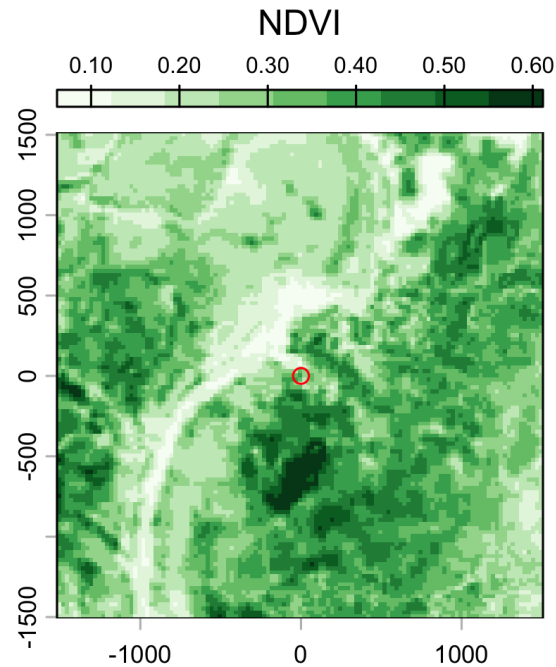
window_extent <- ext(buffalo_single_point$x_ - buffer,
                     buffalo_single_point$x_ + buffer,
                     buffalo_single_point$y_ - buffer,
                     buffalo_single_point$y_ + buffer)

ndvi_window <- crop(ndvi, window_extent)

# set the extent to 0 at the buffalo location
ndvi_window <- terra::shift(ndvi_window,
                           dx = -buffalo_single_point$x_,
                           dy = -buffalo_single_point$y_)

# plot the NDVI layer with the buffalo location as the centre point
plot(ndvi_window,
     # main = paste0("NDVI"),
     col = brewer.pal(9, "Greens"),
     plg=list(x="top", title="NDVI", tics="out", cex=.8))
points(x = 0, y = 0, col = "red")

```



```
# Save the plot
png(filename = paste0(plot_save_path, "/ndvi_local_", which_buffalo, ".png"),
     width = 100, height = 100, units = "mm", res = 600)
plot(ndvi_window,
     # main = paste0("NDVI"),
     col = brewer.pal(9, "Greens"),
     plg=list(x="top", title="NDVI", tics="out", cex=.8))
points(x = 0, y = 0, col = "red")
dev.off()
```

```
quartz_off_screen
2
```

Sample random steps to fit models

```
buffalo_id_steps <- buffalo_id %>%
  steps()

# fitting step length and turning angle distributions to all locations
gamma_dist <- fit_distr(buffalo_id_steps$sl_, "gamma")
vonmises_dist <- fit_distr(buffalo_id_steps$ta_, "vonmises")
```

```
# movement parameters
gamma_dist$params$shape
```

```
[1] 0.4513375
```

```
gamma_dist$params$scale
```

```
[1] 621.3796
```

```
vonmises_dist$params$kappa
```

```
[1] 0.1531606
```

```
vonmises_dist$params$mu
```

```
Circular Data:
```

```
Type = angles
```

```
Units = radians
```

```
Template = none
```

```
Modulo = asis
```

```
Zero = 0
```

```
Rotation = counter
```

```
[1] 0
```

```
# sample random steps
```

```
buffalo_ssf_data <- buffalo_id_steps %>%
```

```
  random_steps(n = 10,
```

```
              sl_dist = gamma_dist,
```

```
              ta_dist = vonmises_dist) %>%
```

```
  extract_covariates(ndvi) %>%
```

```
  mutate(ndvi_sq = ndvi^2,
```

```
         log_sl = log(sl_),
```

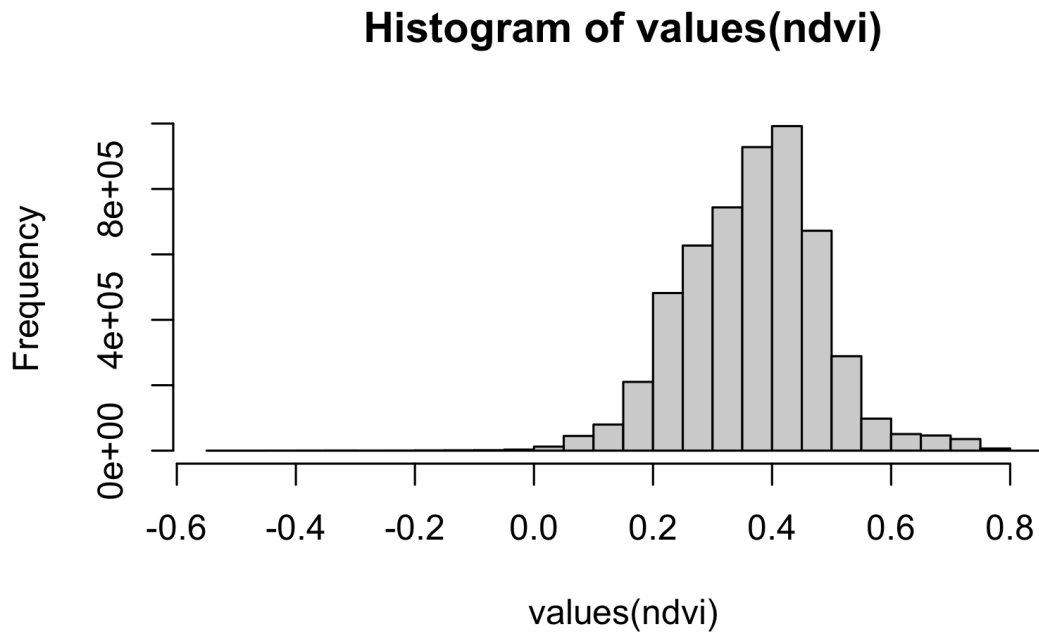
```
         cos_ta = cos(ta_),
```

```
         times = 1) # create a dummy column for times (used in a Cox PH model, but not relevant)
```

Different SSF formulations

Firstly, we want to pull out the range of NDVI values to plot the curves with.

```
# histogram of NDVI values
hist(values(ndvi))
```



```
# 1% and 99% quantiles of NDVI values
ndvi_quantiles <- quantile(values(ndvi), probs = c(0.005, 0.995), na.rm = TRUE)
ndvi_quantiles
```

```
      0.5%      99.5%
0.06121541 0.71819625
```

```
# plot across the range in the local window
ndvi_values <- seq(min(0, ndvi_quantiles[1]),
                  ndvi_quantiles[2],
                  length.out = 100)

pred_data <- data.frame(ndvi = ndvi_values,
                       ndvi_sq = ndvi_values^2,
                       sl_ = 0,
                       log_sl = 0,
                       cos_ta = 0)
```

Set RSS min and max values

This is hard-coded as the maximum rss value across the different formulations, to ensure the plotting extent is consistent across approaches.

```
rss_min <- 0
rss_max <- 2.1
```

Linear covariate

Model formula

$y \sim ndvi$

Fit model

```
# fit an issf model
ssf_linear <- amt::fit_issf(case_ ~
                           ndvi +
                           sl_ + log_sl + cos_ta +
                           strata(step_id_),
                           data = buffalo_ssf_data)

summary(ssf_linear)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + sl_ + log_sl +
      cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)	
ndvi	-8.813e-01	4.143e-01	1.443e-01	-6.105	1.03e-09	***
sl_	1.147e-06	1.000e+00	3.174e-05	0.036	0.971	
log_sl	-2.394e-04	9.998e-01	5.688e-03	-0.042	0.966	
cos_ta	-4.305e-03	9.957e-01	1.503e-02	-0.287	0.774	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.4143	2.414	0.3122	0.5497
sl_	1.0000	1.000	0.9999	1.0001

log_sl	0.9998	1.000	0.9887	1.0110
cos_ta	0.9957	1.004	0.9668	1.0255

Concordance= 0.494 (se = 0.004)
Likelihood ratio test= 38.92 on 4 df, p=7e-08
Wald test = 37.62 on 4 df, p=1e-07
Score (logrank) test = 37.62 on 4 df, p=1e-07

```
AIC(ssf_linear)
```

```
[1] 46478.66
```

```
# fit an ssf gam model with parametric terms
ssf_linear_gam <- mgcv::gam(cbind(times, step_id_) ~
                           ndvi +
                           sl_ + log_sl + cos_ta,
                           data = buffalo_ssf_data,
                           family = cox.ph,
                           weight = case_)

summary(ssf_linear_gam)
```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	-8.813e-01	1.443e-01	-6.105	1.03e-09 ***
sl_	1.147e-06	3.174e-05	0.036	0.971
log_sl	-2.394e-04	5.688e-03	-0.042	0.966
cos_ta	-4.305e-03	1.503e-02	-0.287	0.774

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0544%

-REML = 23254 Scale est. = 1

n = 106678

```
AIC(ssf_linear_gam)
```

```
[1] 46478.66
```

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_linear_gam,
                      newdata = pred_data,
                      type = "link", # log-scale
                      se.fit = TRUE)

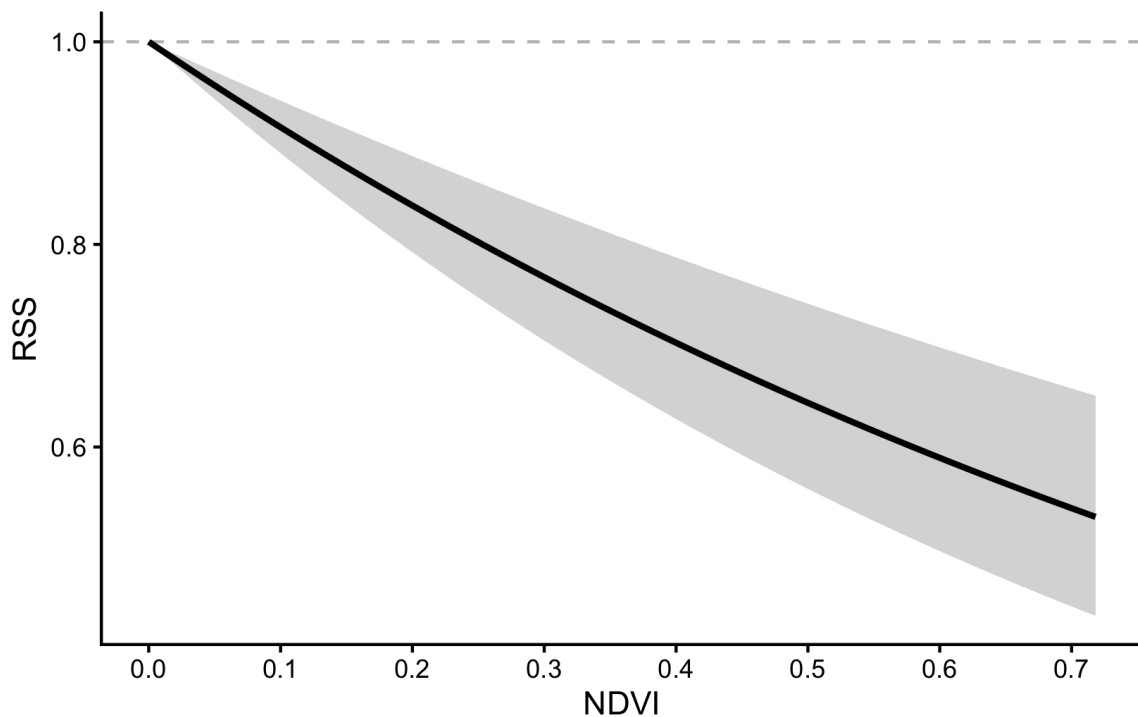
# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
            fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
            colour = "black", linewidth = 1) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()
```



Response curve

```
# extract the NDVI coefficient and its standard error
ndvi_coef <- ssf_linear$model$coefficients[1]
ndvi_coef_se <- sqrt(diag(ssf_linear$model$var))[1] # SE of coefficient

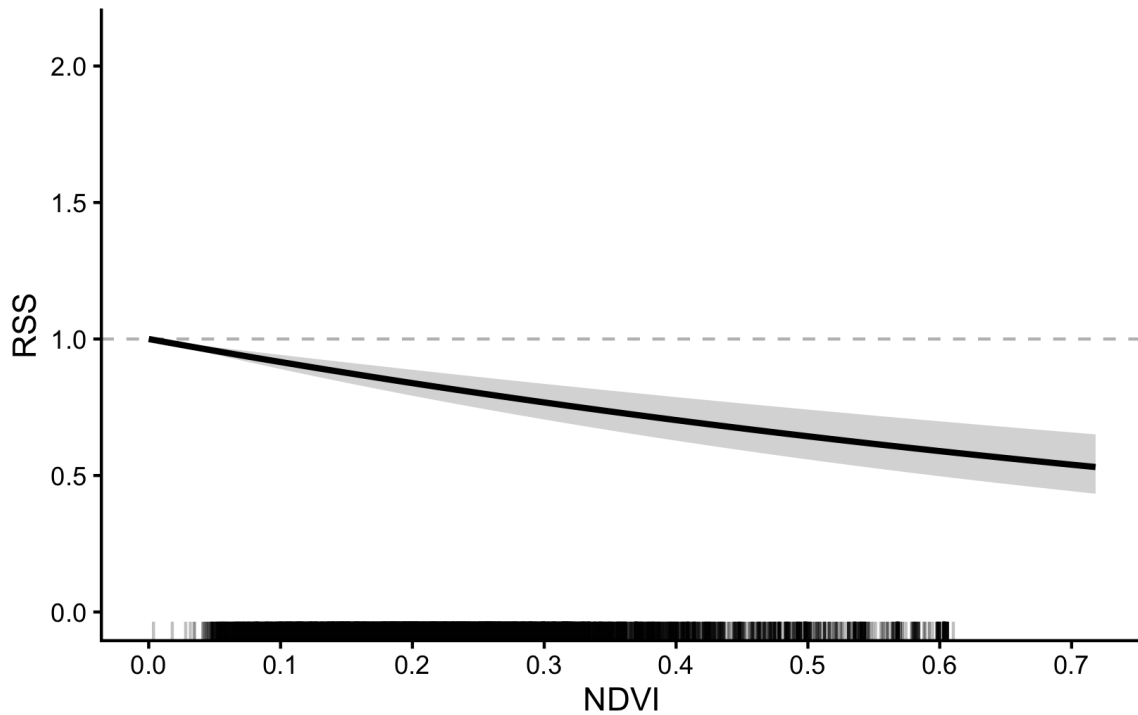
# calculate the RSS for each NDVI value
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = ndvi_values * ndvi_coef,
  rss = exp(ndvi_values * ndvi_coef),
  se_prediction = abs(ndvi_values) * ndvi_coef_se # SE of prediction at each NDVI value
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = exp(log_rss - 1.96 * se_prediction), ymax = exp(log_rss + 1.96 * se_prediction)),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
```

```

aes(x = ndvi), alpha = 0.25, sides="b") +
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
scale_y_continuous(limits = c(rss_min, rss_max)) +
labs(x = "NDVI", y = "RSS") +
theme_classic()

```



```

ggsave(paste0(plot_save_path, "/ndvi_linear_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)

```

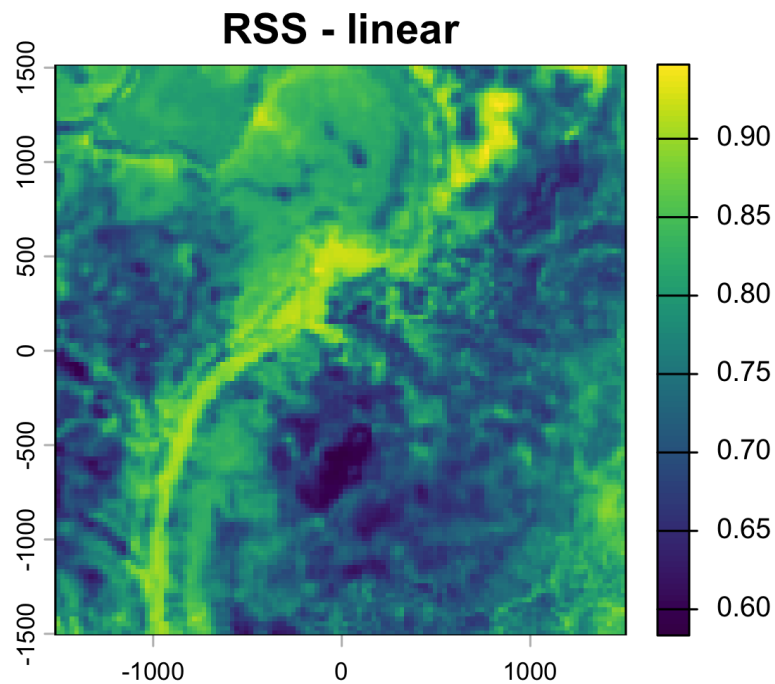
Plot habitat selection

```

# calculate the RSS for the local NDVI values
ndvi_linear <- exp(ndvi_window * ndvi_coef)

# plot the habitat selection (spatial RSS)
plot(ndvi_linear, main = "RSS - linear")

```



```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_linear_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_linear, main = "RSS - linear", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Quadratic covariate

Model formula

$$y \sim ndvi + ndvi^2$$

Fit model

```
# fit an issf model
ssf_quadratic <- amt::fit_issf(case_ ~
                               ndvi +
```

```

        ndvi_sq +
        sl_ + log_sl + cos_ta +
        strata(step_id_),
        data = buffalo_ssf_data)

summary(ssf_quadratic)

```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi_sq +
      sl_ + log_sl + cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.455e+00	8.602e+01	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	1.332e-04	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	1.000e+00	3.183e-05	-0.506	0.613
log_sl	3.391e-03	1.003e+00	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	9.958e-01	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.602e+01	1.163e-02	3.245e+01	2.280e+02
ndvi_sq	1.332e-04	7.510e+03	2.711e-05	6.542e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.003e+00	9.966e-01	9.922e-01	1.015e+00
cos_ta	9.958e-01	1.004e+00	9.668e-01	1.026e+00

Concordance= 0.53 (se = 0.004)

Likelihood ratio test= 171 on 5 df, p=<2e-16

Wald test = 142.3 on 5 df, p=<2e-16

Score (logrank) test = 142.2 on 5 df, p=<2e-16

```
AIC(ssf_quadratic)
```

```
[1] 46348.59
```

```

# fit an ssf gam model with parametric terms. -ends up being the same as above,
# but allows us to use the `predict` function to get predictions on the log-scale with stand
# which is a bit more work to do with the `amt` model object.
ssf_quadratic_gam <- mgcv::gam(cbind(times, step_id_) ~
                                ndvi +

```

```

        ndvi_sq +
        sl_ + log_sl + cos_ta,
        data = buffalo_ssf_data,
        family = cox.ph,
        weight = case_)

summary(ssf_quadratic_gam)

```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + ndvi_sq + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	4.455e+00	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	3.183e-05	-0.506	0.613
log_sl	3.391e-03	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0158%

-REML = 23187 Scale est. = 1 n = 106678

```
AIC(ssf_quadratic_gam)
```

```
[1] 46348.59
```

Generate the response curve

```

# Get predictions on log-scale with standard errors
predictions <- predict(ssf_quadratic_gam,
                      newdata = pred_data,
                      type = "link", # log-scale
                      se.fit = TRUE)

```

```

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

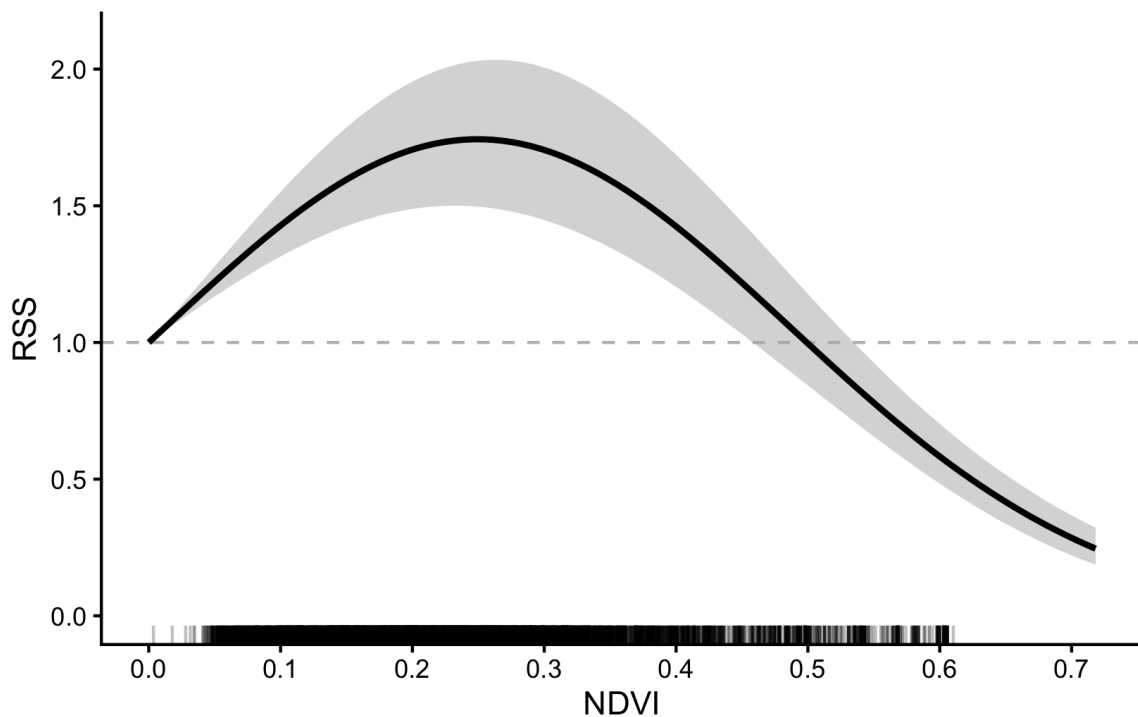
# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(rss_min, rss_max)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()

```



```
ggsave(paste0(plot_save_path, "/ndvi_quadratic_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)
```

Response curve

```
# extract the NDVI coefficient
ndvi_coef <- ssf_quadratic$model$coefficients[1]
ndvi_coef_sq <- ssf_quadratic$model$coefficients[2]

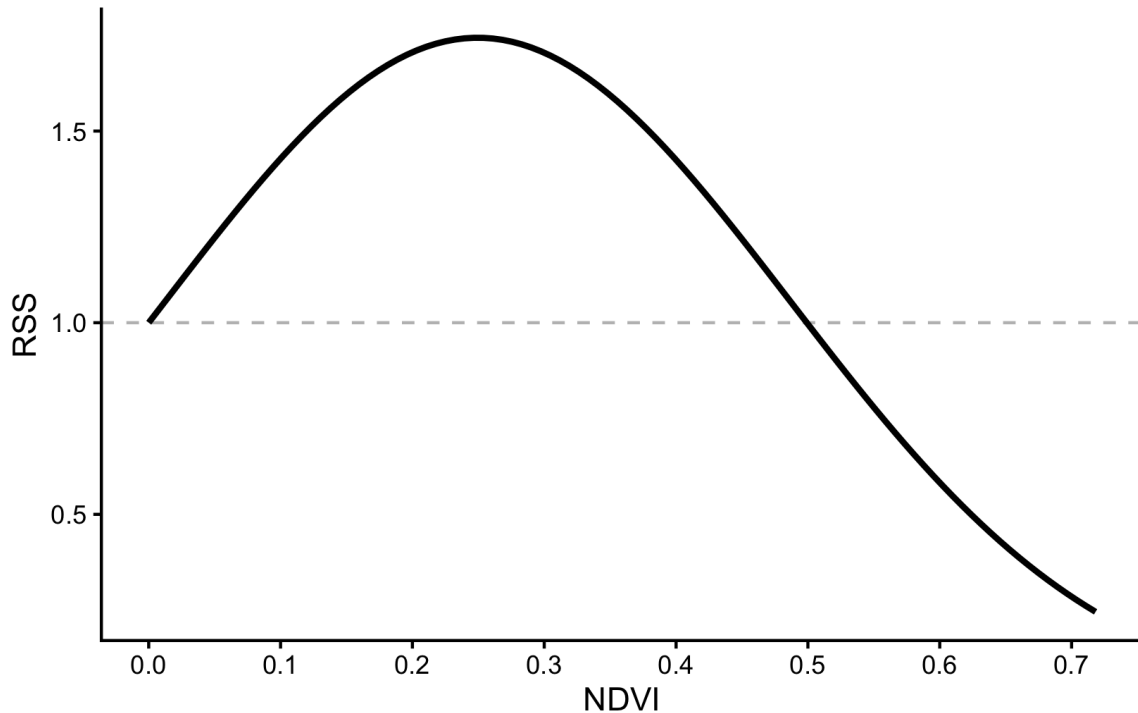
ndvi_se = sqrt(diag(ssf_quadratic$model$var))[1]
ndvi_sq_se = sqrt(diag(ssf_quadratic$model$var))[2]

log_rss = (ndvi_values * ndvi_coef) + (ndvi_values^2 * ndvi_coef_sq)
rss = exp(log_rss)

# calculate the RSS for each NDVI value
ndvi_coef_df <- data.frame(ndvi_values, log_rss, rss)

# plot the response curve
ggplot() +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
```

```
geom_line(data = ndvi_coef_df, aes(x = ndvi_values, y = rss),
          colour = "black", linewidth = 1) +
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
labs(x = "NDVI", y = "RSS") +
theme_classic()
```



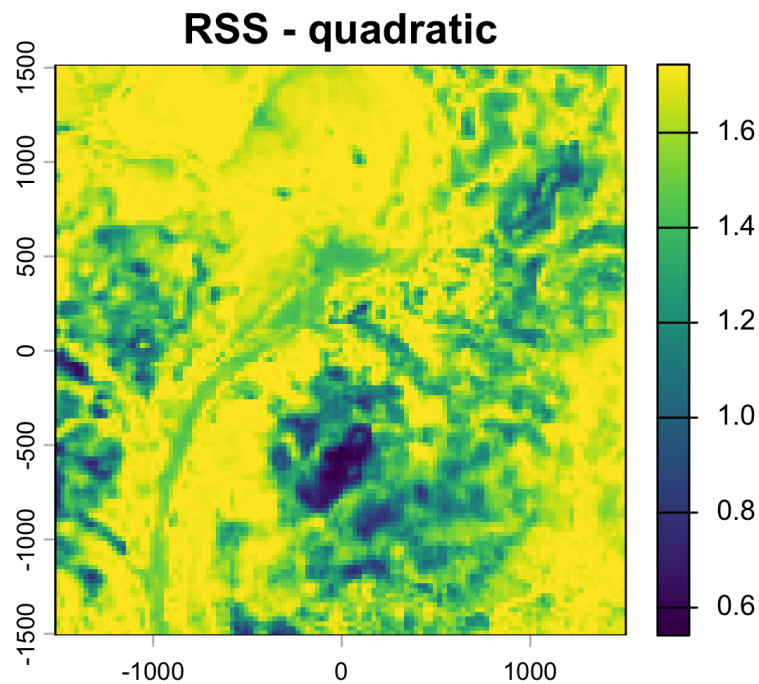
```
# ggsave(paste0(plot_save_path, /ndvi_quadratic_response_id", which_buffalo, ".png"),
#         width = 100, height = 100, units = "mm", dpi = 600)
```

Plot habitat selection

```
# extract the NDVI coefficient
ndvi_coef <- ssf_quadratic$model$coefficients[1]
ndvi_coef_sq <- ssf_quadratic$model$coefficients[2]

# calculate the RSS for the local NDVI values
ndvi_quadratic <- exp((ndvi_window * ndvi_coef) +
                      (ndvi_window^2 * ndvi_coef_sq))

# plot the habitat selection (spatial RSS)
plot(ndvi_quadratic, main = "RSS - quadratic")
```



```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_quadratic_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_quadratic, main = "RSS - quadratic", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Smooth term

Fit a GAM to the data, with a smooth term across the range of NDVI values (Klappstein et al. 2024).

Model formula

$$y \sim s(ndvi)$$

Fit model

```
# fit an ssf gam model
ssf_gam <- mgcv::gam(cbind(times, step_id_) ~
  s(ndvi, bs = "cr") +
  # s(ndvi) +
  sl_ + log_sl + cos_ta,
  data = buffalo_ssf_data,
  family = cox.ph,
  weight = case_)

summary(ssf_gam)
```

Family: Cox PH

Link function: identity

Formula:

```
cbind(times, step_id_) ~ s(ndvi, bs = "cr") + sl_ + log_sl +
  cos_ta
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-4.585e-06	3.181e-05	-0.144	0.885
log_sl	8.339e-03	5.769e-03	1.445	0.148
cos_ta	-5.143e-03	1.505e-02	-0.342	0.733

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(ndvi)	8.448	8.863	386.8	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = 0.397%

-REML = 23088 Scale est. = 1 n = 106678

```
AIC(ssf_gam)
```

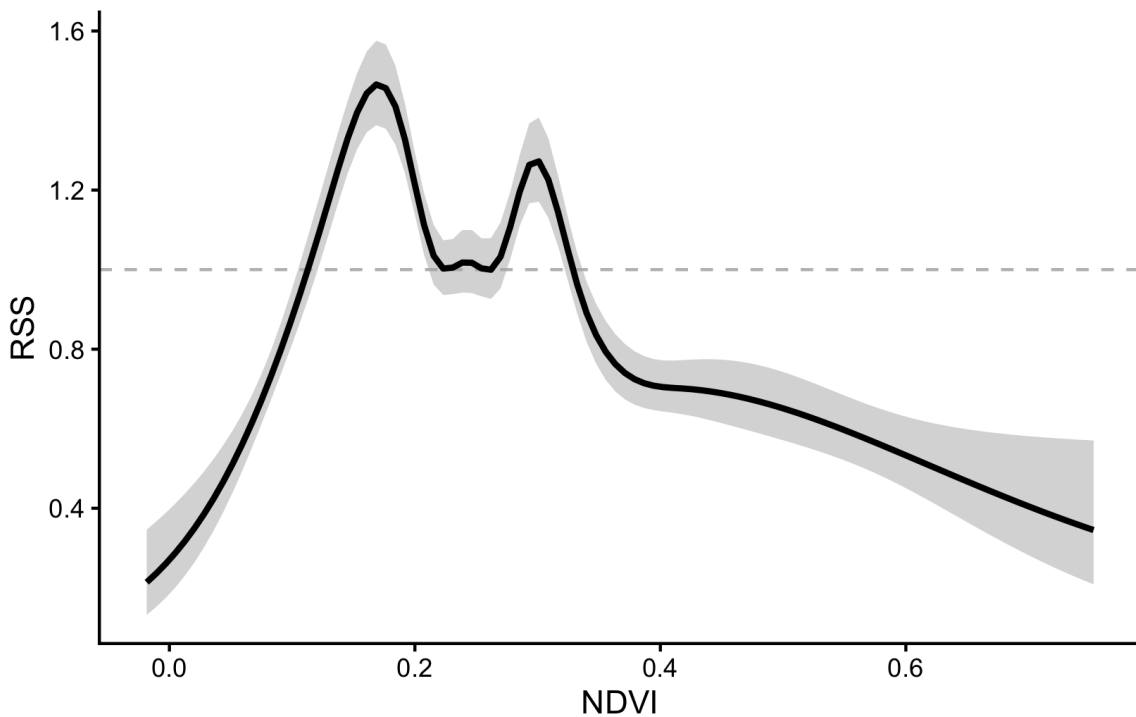
```
[1] 46119.42
```

```
# coef(ssf_gam)[4]
```

```
gam_smooths <- gratia::smooth_estimates(ssf_gam)
```

Using ggplot with the `smooth_estimates` dataframe from `gratia`

```
ggplot(data = gam_smooths) +  
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +  
  geom_ribbon(aes(x = ndvi, ymin = exp(.estimate - 1.96 * .se), ymax = exp(.estimate + 1.96 *  
    .se), fill = "grey70", alpha = 0.5) +  
  geom_line(aes(x = ndvi, y = exp(.estimate)),  
    colour = "black", linewidth = 1) +  
  labs(x = "NDVI", y = "RSS") +  
  # scale_x_continuous(limits = c(ndvi_quantiles[1], 0.5)) +  
  # scale_y_continuous(limits = c(0.65, 1.3)) +  
  theme_classic()
```



```
ggsave(paste0(plot_save_path, "/ndvi_smooth_response_id", which_buffalo, ".png"),  
  width = 120, height = 100, units = "mm", dpi = 600)
```

Alternatively, generate the response curve with the `predict` function

```
# Get predictions on log-scale with standard errors  
predictions <- predict(ssf_gam,  
  newdata = pred_data,
```

```

        type = "link", # log-scale
        se.fit = TRUE)

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

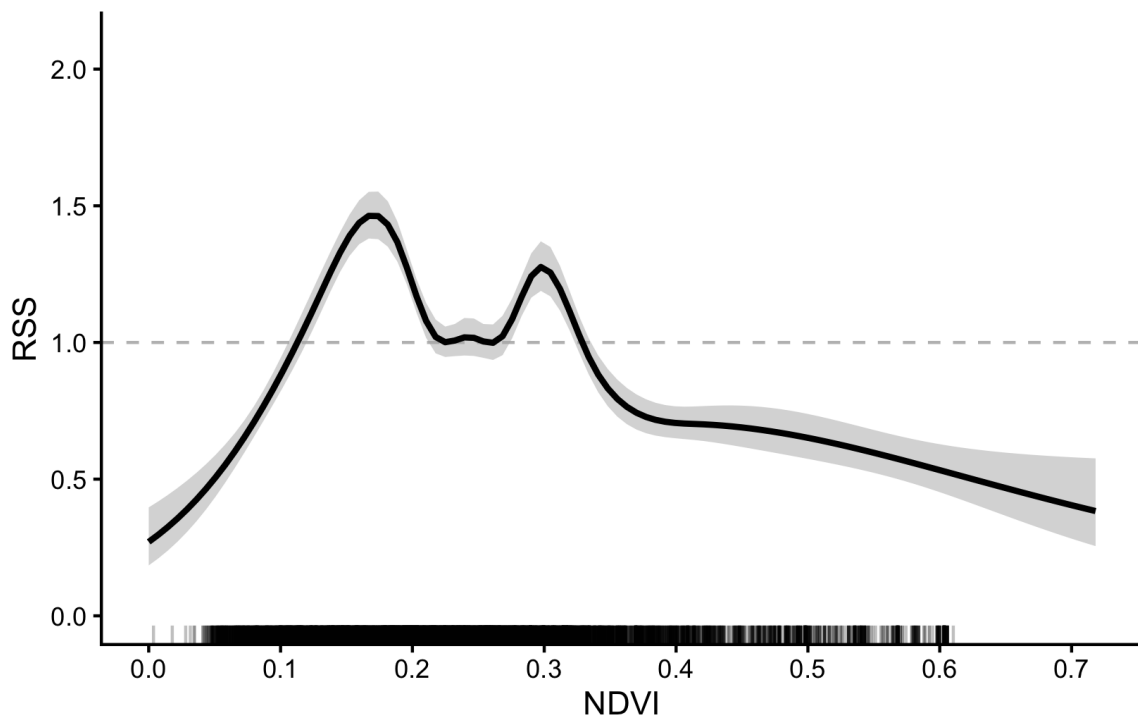
# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper),
    fill = "grey70", alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss),
    colour = "black", linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(rss_min, rss_max)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic()

```



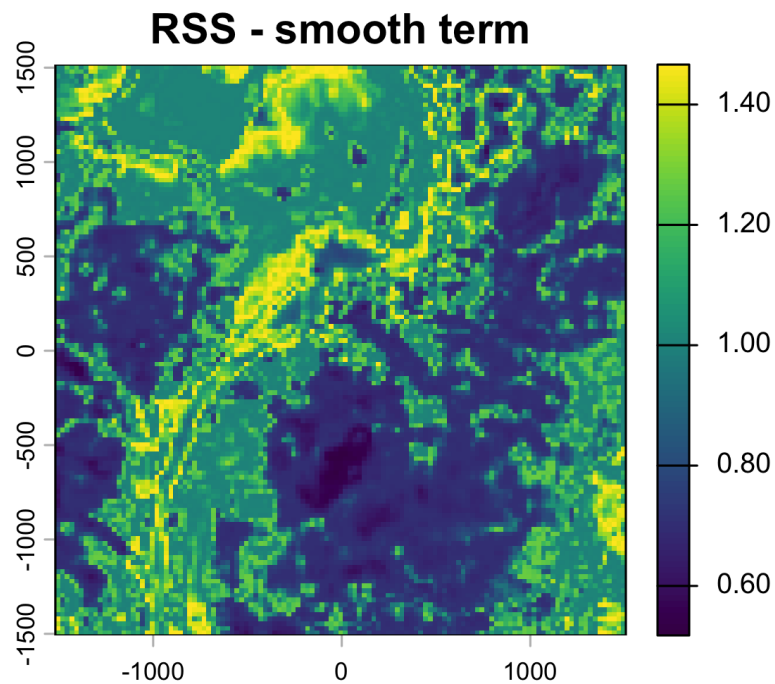
```
ggsave(paste0(plot_save_path, "/ndvi_smooth_response_id", which_buffalo, ".png"),
       width = 120, height = 100, units = "mm", dpi = 600)
```

Plot habitat selection

```
predicted_log_rss <- predict(ssf_gam,
                           newdata = data.frame(ndvi = values(ndvi_window),
                                                  sl_ = 0,
                                                  log_sl = 0,
                                                  cos_ta = 0),,
                           type = "link")

ndvi_gam <- ndvi_window
values(ndvi_gam) <- exp(as.vector(predicted_log_rss))

# Plot the predictions
plot(ndvi_gam, main = "RSS - smooth term")
```



```
# Save the plot
png(filename = paste0(plot_save_path, "/rss_smooth_", which_buffalo, ".png"),
     width = 100, height = 80, units = "mm", res = 600)
plot(ndvi_gam, main = "RSS - smooth", range = c(0.5, 1.75))
dev.off()
```

```
quartz_off_screen
2
```

Temporal dynamics

Static model for comparison

```
ssf_static <- amt::fit_issf(case_ ~
                           ndvi + sl_ + log_sl + cos_ta +
                           strata(step_id_),
                           data = buffalo_ssf_data)

summary(ssf_static)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + sl_ + log_sl +  
      cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	-8.813e-01	4.143e-01	1.443e-01	-6.105	1.03e-09 ***
sl_	1.147e-06	1.000e+00	3.174e-05	0.036	0.971
log_sl	-2.394e-04	9.998e-01	5.688e-03	-0.042	0.966
cos_ta	-4.305e-03	9.957e-01	1.503e-02	-0.287	0.774

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.4143	2.414	0.3122	0.5497
sl_	1.0000	1.000	0.9999	1.0001
log_sl	0.9998	1.000	0.9887	1.0110
cos_ta	0.9957	1.004	0.9668	1.0255

Concordance= 0.494 (se = 0.004)

Likelihood ratio test= 38.92 on 4 df, p=7e-08

Wald test = 37.62 on 4 df, p=1e-07

Score (logrank) test = 37.62 on 4 df, p=1e-07

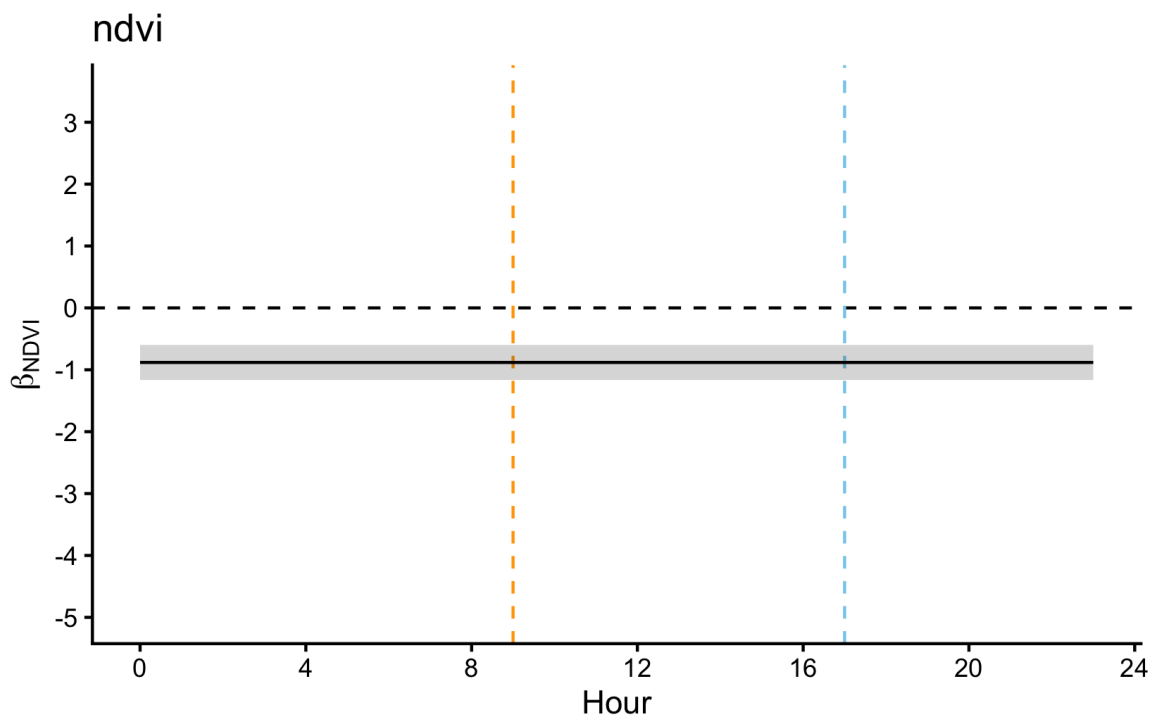
```
AIC(ssf_static)
```

```
[1] 46478.66
```

```
ndvi_coef <- ssf_static$model$coefficients["ndvi"]  
ndvi_se <- sqrt(diag(ssf_static$model$var))[1] # SE of coefficient  
  
ndvi_static_coef_df <- data.frame(hour = seq(0, 23, length.out = 231),  
                                   ndvi_coef = ndvi_coef,  
                                   ndvi_se = ndvi_se,  
                                   ndvi_lo = ndvi_coef - 1.96 * ndvi_se,  
                                   ndvi_hi = ndvi_coef + 1.96 * ndvi_se)
```

Warning in data.frame(hour = seq(0, 23, length.out = 231), ndvi_coef =
ndvi_coef, : row names were found from a short variable and have been discarded

```
# Plot with ribbon
ggplot(ndvi_static_coef_df, aes(x = hour)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  geom_ribbon(aes(ymin = ndvi_lo, ymax = ndvi_hi), alpha = 0.2) +
  geom_line(aes(y = ndvi_coef)) +
  labs(title = "ndvi", x = "Hour", y = expression(beta[NDVI]), fill = NULL) +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  theme_classic()
```



```
ggsave(paste0(plot_save_path, "/ndvi_static_coef_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
sample_hours <- c(9, 17)

ndvi_coef_df <- purrr::map_dfr(sample_hours, function(sample_hour) {
```

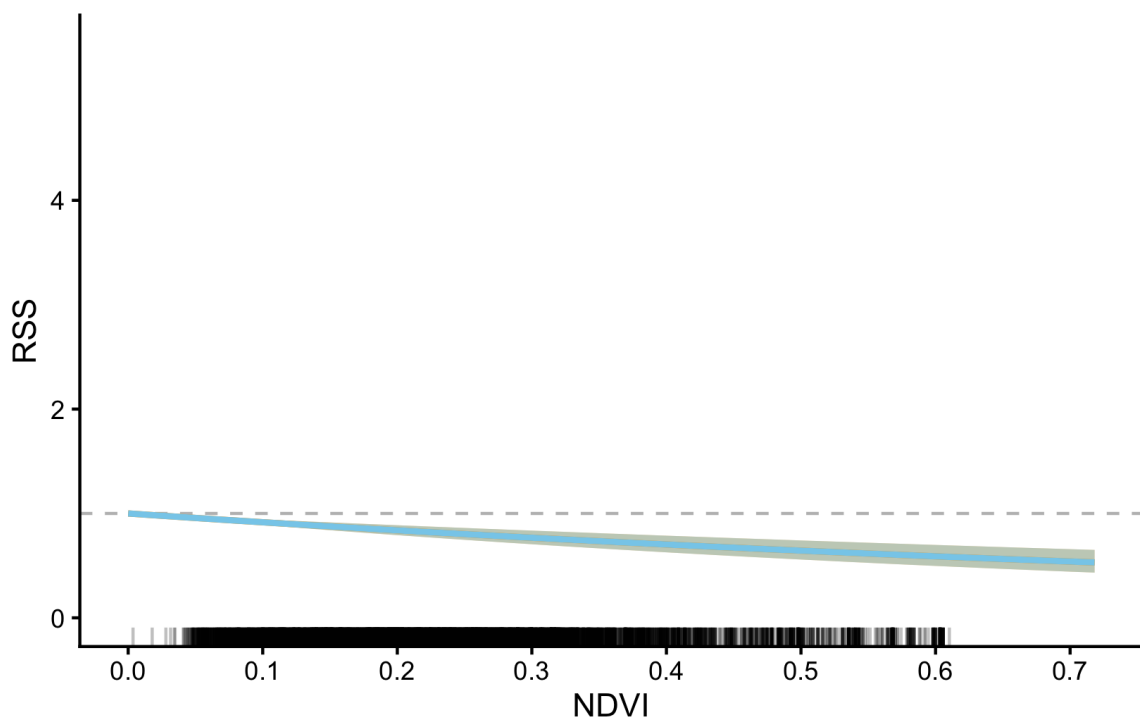
```

# extract the NDVI coefficient and its standard error
ndvi_coef <- ndvi_static_coef_df %>% filter(hour == sample_hour) %>% pull(ndvi_coef)
ndvi_coef_se <- ndvi_static_coef_df %>% filter(hour == sample_hour) %>% pull(ndvi_se)

# calculate the RSS for each NDVI value
data.frame(
  ndvi_values = ndvi_values,
  log_rss = ndvi_values * ndvi_coef,
  rss = exp(ndvi_values * ndvi_coef),
  se_prediction = abs(ndvi_values) * ndvi_coef_se
) %>% mutate(hour = as.factor(sample_hour), .before = ndvi_values)
})

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values,
                ymin = exp(log_rss - 1.96 * se_prediction),
                ymax = exp(log_rss + 1.96 * se_prediction),
                fill = hour),
            alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = hour),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
          aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```



```
ggsave(paste0(plot_save_path, "/ndvi_static_response_sample_hour_id", which_buffalo, ".png"),
       width = 70, height = 80, units = "mm", dpi = 600)
```

Fit harmonic model with quadratic terms

```
ssf_static_quadratic <- amt::fit_issf(case_ ~
  # linear term
  ndvi +
  # quadratic terms
  ndvi_sq +
  sl_ + log_sl + cos_ta +
  strata(step_id_),
  data = buffalo_ssf_data)

summary(ssf_static_quadratic)
```

Call:

```
coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi_sq +
  sl_ + log_sl + cos_ta + strata(step_id_), data = data, method = "exact")
```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.455e+00	8.602e+01	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	1.332e-04	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	1.000e+00	3.183e-05	-0.506	0.613
log_sl	3.391e-03	1.003e+00	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	9.958e-01	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.602e+01	1.163e-02	3.245e+01	2.280e+02
ndvi_sq	1.332e-04	7.510e+03	2.711e-05	6.542e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.003e+00	9.966e-01	9.922e-01	1.015e+00
cos_ta	9.958e-01	1.004e+00	9.668e-01	1.026e+00

Concordance= 0.53 (se = 0.004)
Likelihood ratio test= 171 on 5 df, p=<2e-16
Wald test = 142.3 on 5 df, p=<2e-16
Score (logrank) test = 142.2 on 5 df, p=<2e-16

```
AIC(ssf_static_quadratic)
```

```
[1] 46348.59
```

Create table of NDVI and hour values

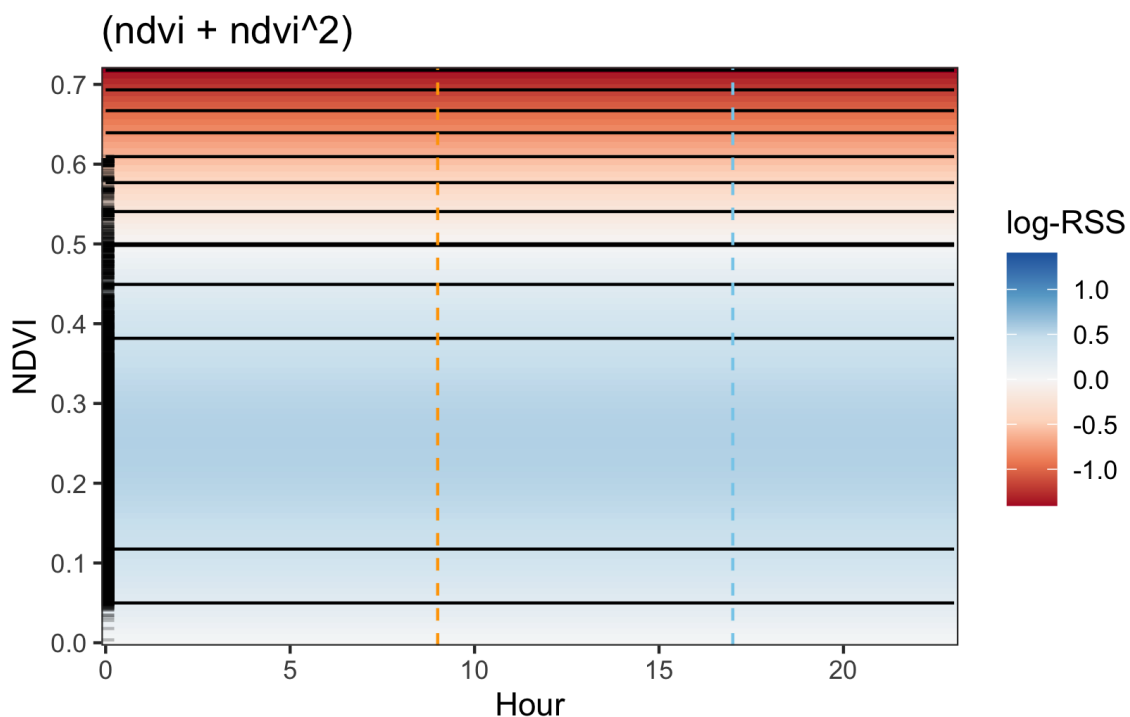
```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                                hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_lin_coef <- ssf_static_quadratic$model$coefficients["ndvi"]
ndvi_quad_coef <- ssf_static_quadratic$model$coefficients["ndvi_sq"]

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  ndvi_lin_coef = ndvi_lin_coef,
  ndvi_quad_coef = ndvi_quad_coef,
  ssf_quadratic_log_rss = (ndvi * ndvi_lin_coef) + (ndvi^2 * ndvi_quad_coef)
)

max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_quadratic_log_rss)))
```

```
# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_quadratic_log_rss)) +
  labs(title = "(ndvi + ndvi^2)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 0.75) +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_static_quadratic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Easier to get the curves with GAM predict function

Fit model

```
# fit an ssf gam model with parametric terms. -ends up being the same as above,
# but allows us to use the `predict` function to get predictions on the log-scale with stand
# which is a bit more work to do with the `amt` model object.
ssf_quadratic_gam <- mgcv::gam(cbind(times, step_id_) ~
                               ndvi +
                               ndvi_sq +
                               sl_ + log_sl + cos_ta,
                               data = buffalo_ssf_data,
                               family = cox.ph,
                               weight = case_)

summary(ssf_quadratic_gam)
```

Family: Cox PH

Link function: identity

Formula:

cbind(times, step_id_) ~ ndvi + ndvi_sq + sl_ + log_sl + cos_ta

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ndvi	4.455e+00	4.973e-01	8.958	<2e-16 ***
ndvi_sq	-8.924e+00	8.122e-01	-10.988	<2e-16 ***
sl_	-1.610e-05	3.183e-05	-0.506	0.613
log_sl	3.391e-03	5.720e-03	0.593	0.553
cos_ta	-4.258e-03	1.504e-02	-0.283	0.777

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -0.0158%

-REML = 23187 Scale est. = 1

n = 106678

```
AIC(ssf_quadratic_gam)
```

```
[1] 46348.59
```

Generate the response curve

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_quadratic_gam,
                      newdata = pred_data,
                      type = "link", # log-scale
                      se.fit = TRUE)

# Extract log RSS and standard errors
log_rss <- predictions$fit
se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
log_rss_lower <- log_rss - 1.96 * se_log_rss
log_rss_upper <- log_rss + 1.96 * se_log_rss

# Transform to RSS scale
rss <- exp(log_rss)
rss_lower <- exp(log_rss_lower)
rss_upper <- exp(log_rss_upper)

# Create your dataframe
ndvi_coef_df <- data.frame(
  ndvi_values = ndvi_values,
  log_rss = log_rss,
  log_rss_lower = log_rss_lower,
  log_rss_upper = log_rss_upper,
  rss = rss,
  rss_lower = rss_lower,
  rss_upper = rss_upper
)

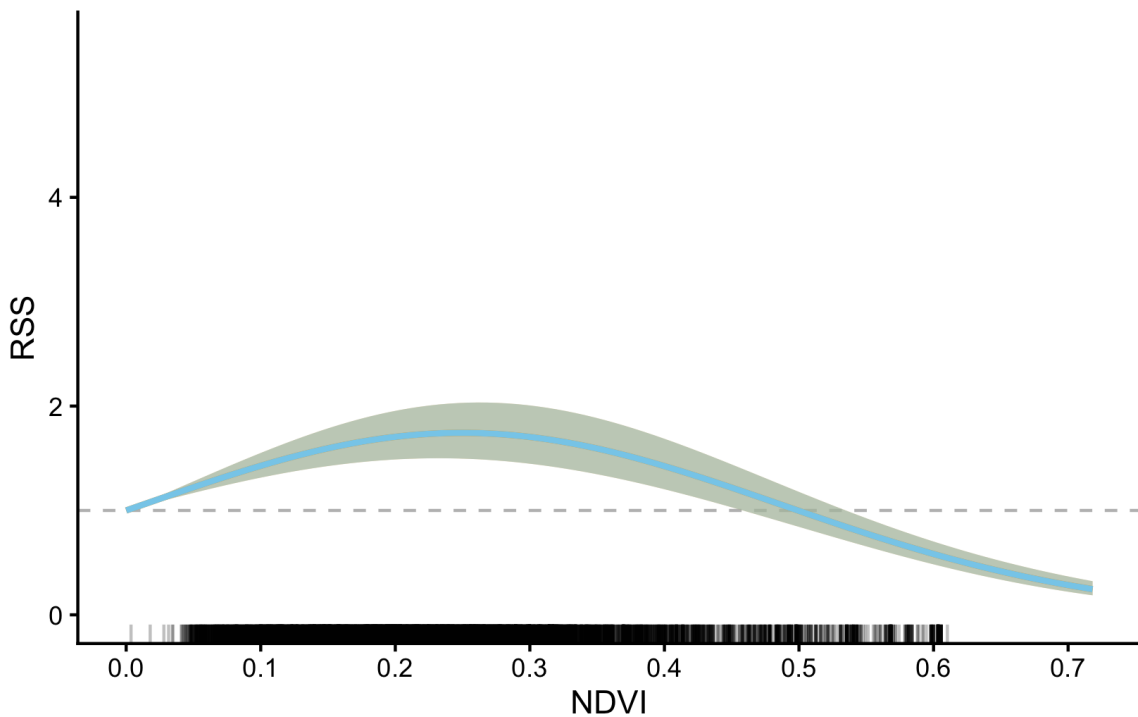
ndvi_coef_df <- rbind(data.frame(hour = 9, ndvi_coef_df), data.frame(hour = 17, ndvi_coef_df))

# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values, ymin = rss_lower, ymax = rss_upper, fill = as.factor(hour))
```

```

    alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = as.factor(hour)),
    linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```



```

ggsave(paste0(plot_save_path, "/ndvi_static_quadratic_response_sample_hour_id", which_buffal
  width = 70, height = 80, units = "mm", dpi = 600)

```

Harmonic regression

```

buffalo_ssf_data <- buffalo_ssf_data %>% mutate(

```

```

# create hour term
hour = lubridate::hour(t1_),

# harmonic terms
sin_1_hour = sin(2 * pi * hour / 24),
cos_1_hour = cos(2 * pi * hour / 24),
sin_2_hour = sin(4 * pi * hour / 24),
cos_2_hour = cos(4 * pi * hour / 24),

)

```

Fit model

We'll use the `fit_issf` function from `amt`, with manually specified harmonic terms.

```

ssf_harmonics <- amt::fit_issf(case_ ~
                                ndvi +
                                ndvi:sin_1_hour + ndvi:cos_1_hour +
                                ndvi:sin_2_hour + ndvi:cos_2_hour +
                                sl_ + log_sl + cos_ta +
                                strata(step_id_),
                                data = buffalo_ssf_data)

summary(ssf_harmonics)

```

Call:

```

coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi:sin_1_hour +
        ndvi:cos_1_hour + ndvi:sin_2_hour + ndvi:cos_2_hour + sl_ +
        log_sl + cos_ta + strata(step_id_), data = data, method = "exact")

```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	-9.534e-01	3.854e-01	1.474e-01	-6.467	9.97e-11 ***
sl_	-9.277e-06	1.000e+00	3.189e-05	-0.291	0.77113
log_sl	4.017e-04	1.000e+00	5.699e-03	0.070	0.94381
cos_ta	-4.982e-03	9.950e-01	1.503e-02	-0.331	0.74036
ndvi:sin_1_hour	1.195e+00	3.304e+00	2.165e-01	5.520	3.38e-08 ***
ndvi:cos_1_hour	-7.601e-01	4.676e-01	1.976e-01	-3.846	0.00012 ***
ndvi:sin_2_hour	-1.063e+00	3.453e-01	2.066e-01	-5.148	2.64e-07 ***
ndvi:cos_2_hour	8.069e-01	2.241e+00	2.074e-01	3.891	9.97e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	0.3854	2.5946	0.2887	0.5145
sl_	1.0000	1.0000	0.9999	1.0001
log_sl	1.0004	0.9996	0.9893	1.0116
cos_ta	0.9950	1.0050	0.9661	1.0248
ndvi:sin_1_hour	3.3036	0.3027	2.1613	5.0495
ndvi:cos_1_hour	0.4676	2.1385	0.3174	0.6889
ndvi:sin_2_hour	0.3453	2.8963	0.2303	0.5176
ndvi:cos_2_hour	2.2410	0.4462	1.4925	3.3648

Concordance= 0.521 (se = 0.004)

Likelihood ratio test= 122.8 on 8 df, p=<2e-16

Wald test = 115.8 on 8 df, p=<2e-16

Score (logrank) test = 115 on 8 df, p=<2e-16

```
AIC(ssf_harmonics)
```

```
[1] 46402.8
```

Pull out the coefficient across time

```
ndvi_harmonic_coef_reconstruction <- function(ssf_harmonic_model, hour) {

  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]

  sin_1_hour <- sin(2 * pi * hour / 24)
  cos_1_hour <- cos(2 * pi * hour / 24)
  sin_2_hour <- sin(4 * pi * hour / 24)
  cos_2_hour <- cos(4 * pi * hour / 24)

  coef <- as.numeric(ndvi_coef +
    sin_1_hour * ndvi_sin_1_coef +
    cos_1_hour * ndvi_cos_1_coef +
    sin_2_hour * ndvi_sin_2_coef +
    cos_2_hour * ndvi_cos_2_coef)

  return(coef)
}
```

```

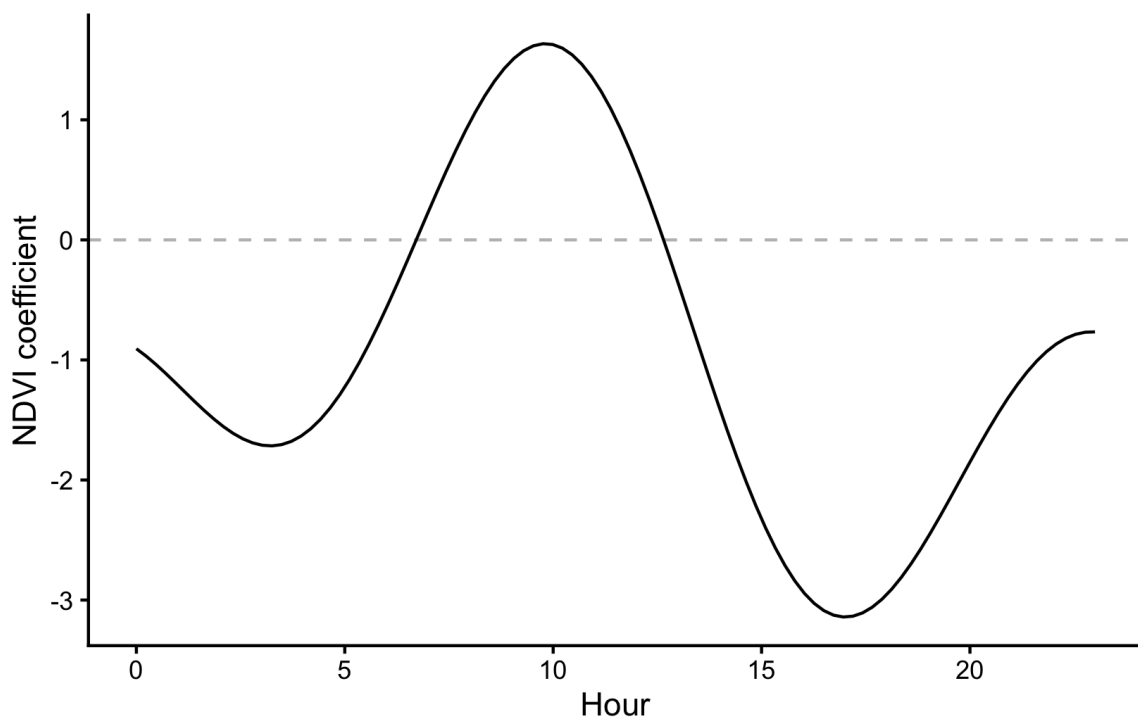
}

ndvi_temporal_coef_df <- data.frame(hour = seq(0, 23, length.out = 100))

ndvi_temporal_coef_df$ndvi_coef <- sapply(ndvi_temporal_coef_df$hour,
                                           ndvi_harmonic_coef_reconstruction,
                                           ssf_harmonic_model = ssf_harmonics)

# plot the response curve
ggplot(data = ndvi_temporal_coef_df) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey") +
  geom_line(aes(x = hour, y = ndvi_coef),
            colour = "black") +
  labs(x = "Hour", y = "NDVI coefficient") +
  theme_classic()

```



Calculate confidence intervals across time and plot

```

sample_hour <- 9 # hour to plot with a vertical dashed line

```

```

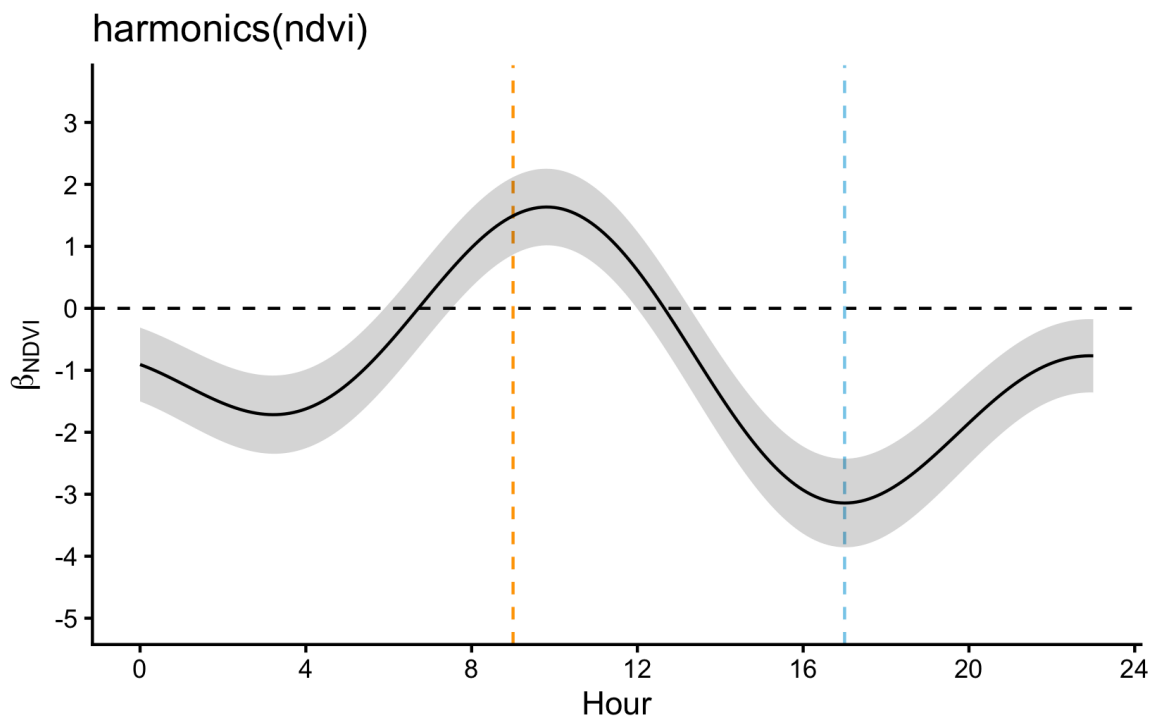
# Coefficient names we need
coef_names <- c("ndvi",
               "ndvi:sin_1_hour", "ndvi:cos_1_hour",
               "ndvi:sin_2_hour", "ndvi:cos_2_hour")

# coefficients and V from the underlying clogit model
beta <- coef(ssf_harmonics$model)[coef_names]
V      <- vcov(ssf_harmonics$model)[coef_names, coef_names, drop = FALSE]

# Build design for each hour
ndvi_temporal_ci <- tibble(hour = seq(0, 23, length.out = 231)) %>%
  mutate(
    sin_1 = sin(2 * pi * hour / 24),
    cos_1 = cos(2 * pi * hour / 24),
    sin_2 = sin(4 * pi * hour / 24),
    cos_2 = cos(4 * pi * hour / 24)
  ) %>%
  rowwise() %>%
  mutate(
    x = list(c(1, sin_1, cos_1, sin_2, cos_2)),
    est = as.numeric(matrix(unlist(x), nrow = 1) %*% beta),
    se  = sqrt(as.numeric(matrix(unlist(x), nrow = 1) %*% V %*% matrix(unlist(x), ncol = 1))),
    lo  = est - 1.96 * se,
    hi  = est + 1.96 * se
  ) %>%
  ungroup() %>%
  select(hour, est, se, lo, hi)

# Plot with ribbon
ggplot(ndvi_temporal_ci, aes(x = hour)) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2) +
  geom_line(aes(y = est)) +
  labs(title = "harmonics(ndvi)", x = "Hour", y = expression(beta[NDVI]), fill = NULL) +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  theme_classic()

```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_coef_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

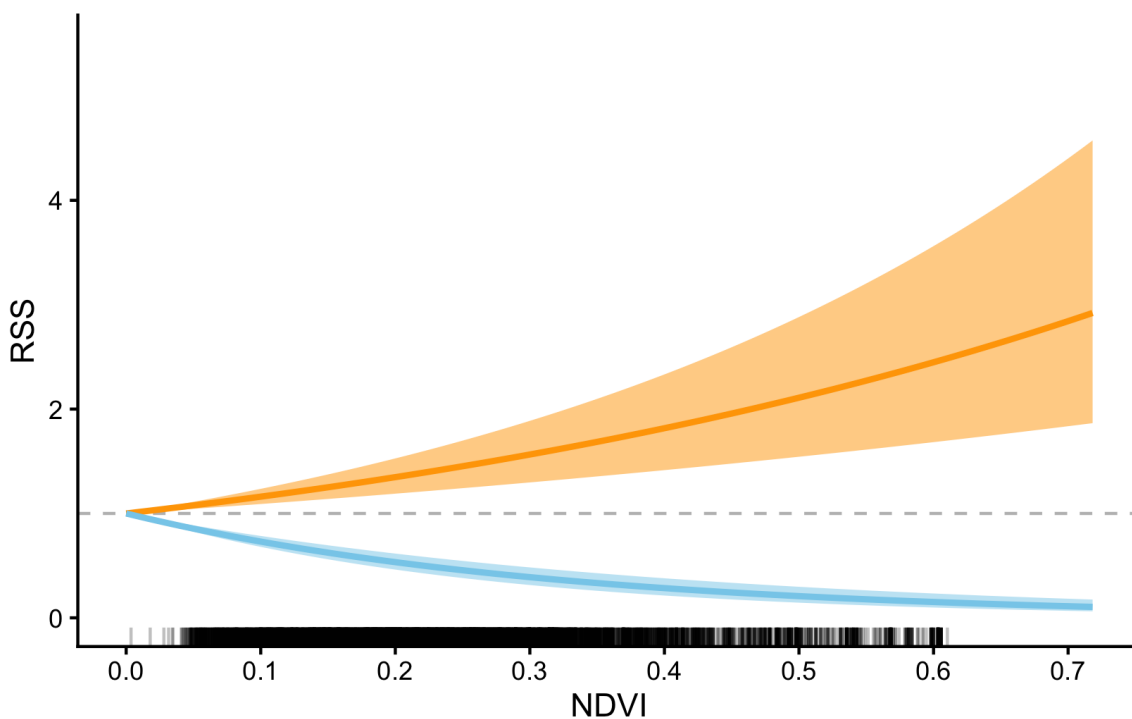
```
sample_hours <- c(9, 17)

ndvi_coef_df <- purrr::map_dfr(sample_hours, function(sample_hour) {

  # extract the NDVI coefficient and its standard error
  ndvi_coef <- ndvi_temporal_ci %>% filter(hour == sample_hour) %>% pull(est)
  ndvi_coef_se <- ndvi_temporal_ci %>% filter(hour == sample_hour) %>% pull(se)

  # calculate the RSS for each NDVI value
  data.frame(
    ndvi_values = ndvi_values,
    log_rss = ndvi_values * ndvi_coef,
    rss = exp(ndvi_values * ndvi_coef),
    se_prediction = abs(ndvi_values) * ndvi_coef_se
  ) %>% mutate(hour = as.factor(sample_hour), .before = ndvi_values)
})
```

```
# plot the response curve
ggplot(data = ndvi_coef_df) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi_values,
                 ymin = exp(log_rss - 1.96 * se_prediction),
                 ymax = exp(log_rss + 1.96 * se_prediction),
                 fill = hour),
            alpha = 0.5) +
  geom_line(aes(x = ndvi_values, y = rss, colour = hour),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
          aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")
```



```
ggsave(paste0(plot_save_path, "/ndvi_linear_response_sample_hour_id", which_buffalo, ".png")
       width = 70, height = 80, units = "mm", dpi = 600)
```

Manually create the selection surface

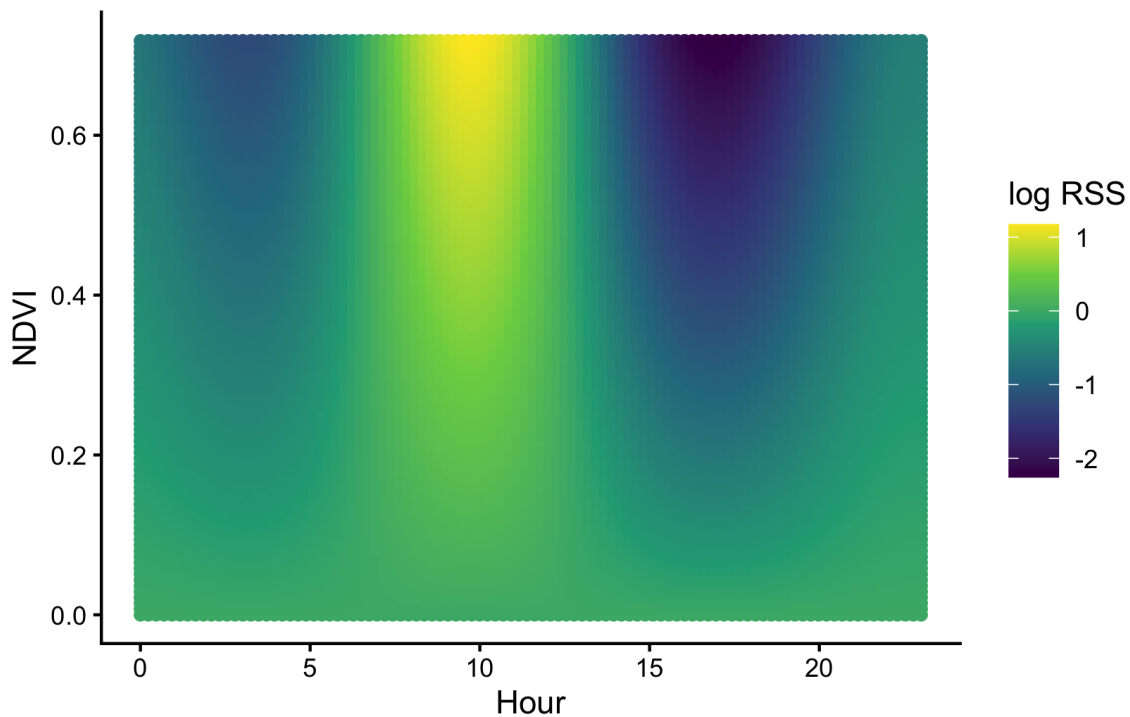
```
ndvi_harmonic_reconstruction <- function(ssf_harmonic_model, ndvi_value, hour) {  
  
  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]  
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]  
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]  
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]  
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]  
  
  sin_1_hour <- sin(2 * pi * hour / 24)  
  cos_1_hour <- cos(2 * pi * hour / 24)  
  sin_2_hour <- sin(4 * pi * hour / 24)  
  cos_2_hour <- cos(4 * pi * hour / 24)  
  
  log_rss <- as.numeric(ndvi_value * ndvi_coef +  
                        ndvi_value * sin_1_hour * ndvi_sin_1_coef +  
                        ndvi_value * cos_1_hour * ndvi_cos_1_coef +  
                        ndvi_value * sin_2_hour * ndvi_sin_2_coef +  
                        ndvi_value * cos_2_hour * ndvi_cos_2_coef)  
  
  return(log_rss)  
}  
  
# test function  
ndvi_harmonic_reconstruction(ssf_harmonics,  
                             ndvi_values[1],  
                             hour = 12)
```

```
[1] 0
```

Create table of NDVI and hour values

```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,  
                               hour = seq(0, 23, length.out = length(ndvi_values)))  
  
ndvi_hour_values$ssf_linear_log_rss <- mapply(ndvi_harmonic_reconstruction,  
                                              ssf_harmonic_model = list(ssf_harmonics),  
                                              ndvi_value = ndvi_hour_values$ndvi,  
                                              hour = ndvi_hour_values$hour)
```

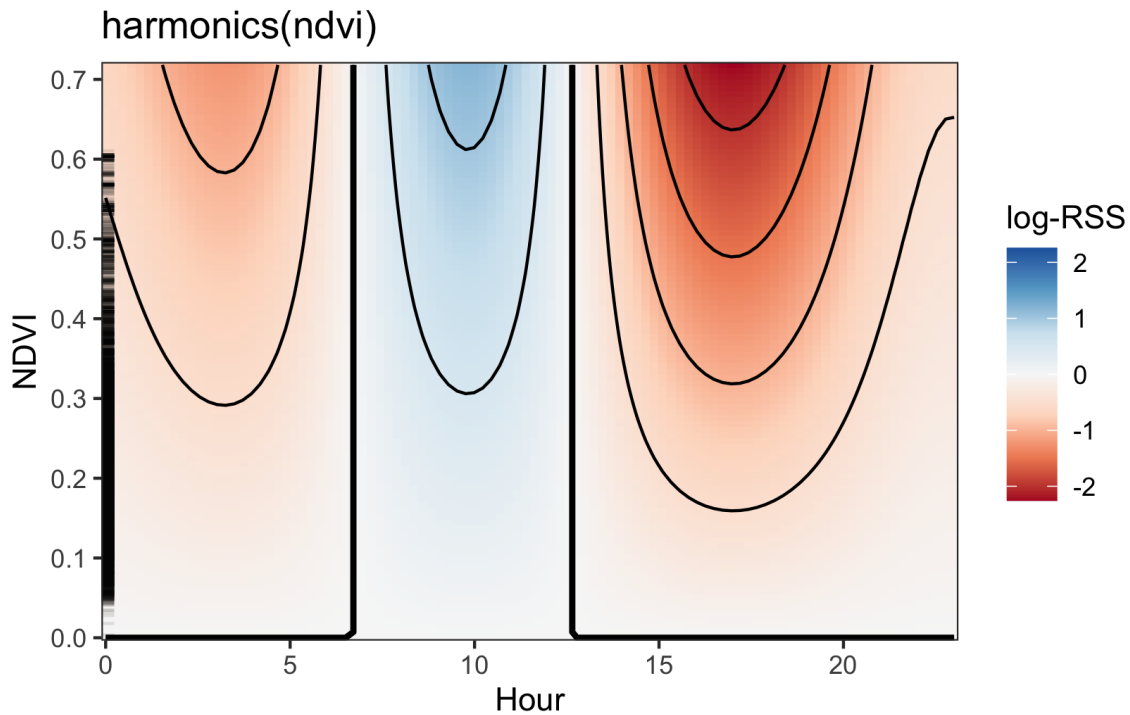
```
# plot the response curve
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = ssf_linear_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  theme_classic()
```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_linear_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_linear_log_rss)) +
  labs(title = "harmonics(ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_linear_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_linear_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 1) +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) + # symmetric around 0
  coord_cartesian(expand = FALSE) +
```

```
geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
         aes(y = ndvi), alpha = 0.1, length = unit(0.015, "npc"), sides="l") +
scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Fit harmonic model with quadratic terms

```
ssf_harmonics_quadratic <- amt::fit_issf(case_ ~

# linear terms
ndvi +
ndvi:sin_1_hour + ndvi:cos_1_hour +
ndvi:sin_2_hour + ndvi:cos_2_hour +

# quadratic terms
ndvi_sq +
ndvi_sq:sin_1_hour + ndvi_sq:cos_1_hour +
```

```

ndvi_sq:sin_2_hour + ndvi_sq:cos_2_hour +

sl_ + log_sl + cos_ta +

strata(step_id_),
data = buffalo_ssf_data)

summary(ssf_harmonics_quadratic)

```

Call:

```

coxph(formula = Surv(rep(1, 106678L), case_) ~ ndvi + ndvi:sin_1_hour +
  ndvi:cos_1_hour + ndvi:sin_2_hour + ndvi:cos_2_hour + ndvi_sq +
  ndvi_sq:sin_1_hour + ndvi_sq:cos_1_hour + ndvi_sq:sin_2_hour +
  ndvi_sq:cos_2_hour + sl_ + log_sl + cos_ta + strata(step_id_),
  data = data, method = "exact")

```

n= 106678, number of events= 9698

	coef	exp(coef)	se(coef)	z	Pr(> z)
ndvi	4.496e+00	8.970e+01	5.067e-01	8.873	< 2e-16 ***
ndvi_sq	-9.138e+00	1.075e-04	8.400e-01	-10.878	< 2e-16 ***
sl_	-2.582e-05	1.000e+00	3.197e-05	-0.808	0.41923
log_sl	4.076e-03	1.004e+00	5.731e-03	0.711	0.47695
cos_ta	-5.770e-03	9.942e-01	1.505e-02	-0.384	0.70134
ndvi:sin_1_hour	8.621e-01	2.368e+00	7.214e-01	1.195	0.23209
ndvi:cos_1_hour	1.219e+00	3.385e+00	7.034e-01	1.734	0.08299 .
ndvi:sin_2_hour	-2.196e+00	1.113e-01	7.186e-01	-3.056	0.00225 **
ndvi:cos_2_hour	1.471e+00	4.353e+00	7.043e-01	2.088	0.03676 *
ndvi_sq:sin_1_hour	6.008e-01	1.824e+00	1.226e+00	0.490	0.62410
ndvi_sq:cos_1_hour	-3.012e+00	4.919e-02	1.126e+00	-2.675	0.00747 **
ndvi_sq:sin_2_hour	1.688e+00	5.411e+00	1.176e+00	1.436	0.15104
ndvi_sq:cos_2_hour	-5.083e-01	6.015e-01	1.174e+00	-0.433	0.66493

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
ndvi	8.970e+01	1.115e-02	3.322e+01	2.422e+02
ndvi_sq	1.075e-04	9.298e+03	2.073e-05	5.580e-04
sl_	1.000e+00	1.000e+00	9.999e-01	1.000e+00
log_sl	1.004e+00	9.959e-01	9.929e-01	1.015e+00
cos_ta	9.942e-01	1.006e+00	9.654e-01	1.024e+00
ndvi:sin_1_hour	2.368e+00	4.223e-01	5.759e-01	9.738e+00
ndvi:cos_1_hour	3.385e+00	2.954e-01	8.528e-01	1.344e+01
ndvi:sin_2_hour	1.113e-01	8.985e+00	2.722e-02	4.551e-01

```

ndvi:cos_2_hour      4.353e+00  2.297e-01  1.095e+00  1.731e+01
ndvi_sq:sin_1_hour  1.824e+00  5.484e-01  1.649e-01  2.016e+01
ndvi_sq:cos_1_hour  4.919e-02  2.033e+01  5.413e-03  4.470e-01
ndvi_sq:sin_2_hour  5.411e+00  1.848e-01  5.400e-01  5.423e+01
ndvi_sq:cos_2_hour  6.015e-01  1.663e+00  6.028e-02  6.002e+00

```

Concordance= 0.549 (se = 0.004)

Likelihood ratio test= 270 on 13 df, p=<2e-16

Wald test = 221.1 on 13 df, p=<2e-16

Score (logrank) test = 221.1 on 13 df, p=<2e-16

```
AIC(ssf_harmonics_quadratic)
```

```
[1] 46265.62
```

Manually create the selection surface

```

ndvi_harmonic_reconstruction <- function(ssf_harmonic_model, ndvi_value, hour) {

  # linear terms
  ndvi_coef <- ssf_harmonic_model$model$coefficients["ndvi"]
  ndvi_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_1_hour"]
  ndvi_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_1_hour"]
  ndvi_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:sin_2_hour"]
  ndvi_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi:cos_2_hour"]

  # quadratic terms
  ndvi_sq_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq"]
  ndvi_sq_sin_1_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:sin_1_hour"]
  ndvi_sq_cos_1_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:cos_1_hour"]
  ndvi_sq_sin_2_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:sin_2_hour"]
  ndvi_sq_cos_2_coef <- ssf_harmonic_model$model$coefficients["ndvi_sq:cos_2_hour"]

  sin_1_hour <- sin(2 * pi * hour / 24)
  cos_1_hour <- cos(2 * pi * hour / 24)
  sin_2_hour <- sin(4 * pi * hour / 24)
  cos_2_hour <- cos(4 * pi * hour / 24)

  linear_part <- ndvi_value * ndvi_coef +
    ndvi_value * sin_1_hour * ndvi_sin_1_coef +
    ndvi_value * cos_1_hour * ndvi_cos_1_coef +
    ndvi_value * sin_2_hour * ndvi_sin_2_coef +

```

```

    ndvi_value * cos_2_hour * ndvi_cos_2_coef

ndvi_sq_value <- ndvi_value^2

quadratic_part <- ndvi_sq_value * ndvi_sq_coef +
  ndvi_sq_value * sin_1_hour * ndvi_sq_sin_1_coef +
  ndvi_sq_value * cos_1_hour * ndvi_sq_cos_1_coef +
  ndvi_sq_value * sin_2_hour * ndvi_sq_sin_2_coef +
  ndvi_sq_value * cos_2_hour * ndvi_sq_cos_2_coef

return(as.numeric(linear_part + quadratic_part))
}

# test function
ndvi_harmonic_reconstruction(ssf_harmonics_quadratic, ndvi_values[1], hour = 12)

```

```
[1] 0
```

Create table of NDVI and hour values

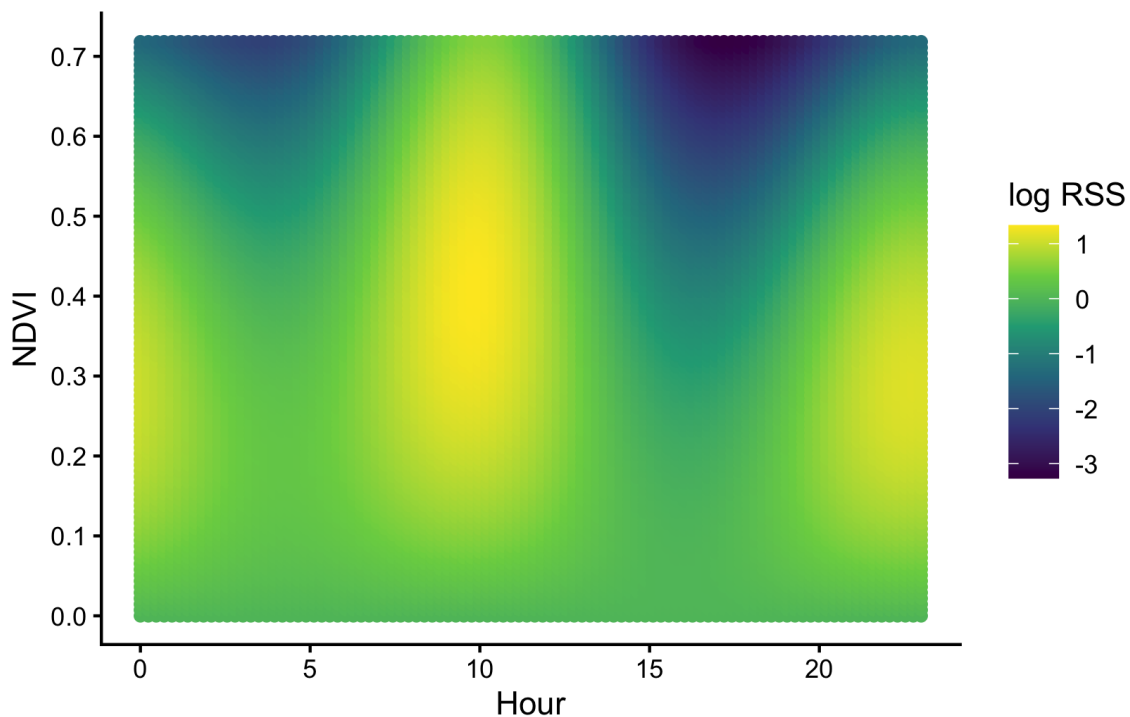
```

ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                                hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values$ssf_quadratic_log_rss <- mapply(ndvi_harmonic_reconstruction,
                                                  ssf_harmonic_model = list(ssf_harmonics_qua
                                                  ndvi_value = ndvi_hour_values$ndvi,
                                                  hour = ndvi_hour_values$hour)

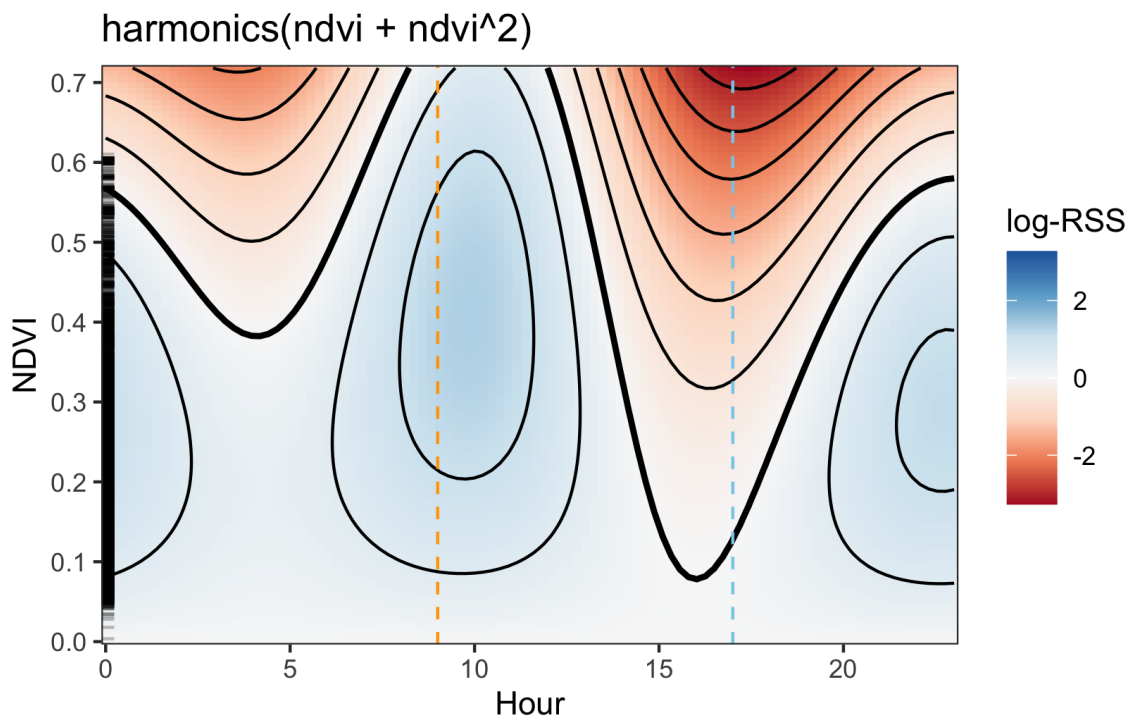
# plot the response curve
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = ssf_quadratic_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_classic()

```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$ssf_quadratic_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = ssf_quadratic_log_rss)) +
  labs(title = "harmonics(ndvi + ndvi^2)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    colour = "black") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = ssf_quadratic_log_rss),
    breaks = 0, colour = "black", linetype = "solid", linewidth = 1) +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_harmonic_quadratic_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
sample_hours <- c(9, 17)

curve_df <- purrr::map_dfr(sample_hours, function(sample_hour) {

  # Harmonics at sample hour
  s1 <- sin(2 * pi * sample_hour / 24)
  c1 <- cos(2 * pi * sample_hour / 24)
  s2 <- sin(4 * pi * sample_hour / 24)
  c2 <- cos(4 * pi * sample_hour / 24)

  mod <- ssf_harmonics_quadratic$model
  betas <- coef(mod)
  V <- vcov(mod)
  cn <- names(betas)

  # For each NDVI value, build the contrast vector g such that g' = log-RSS
```

```

purrr::map_dfr(ndvi_values, function(x) {

  g <- setNames(numeric(length(betas)), cn)

  # linear term:  $x * (_{ndvi} + \text{harmonics})$ 
  g["ndvi"] <- x
  g["ndvi:sin_1_hour"] <- x * s1
  g["ndvi:cos_1_hour"] <- x * c1
  g["ndvi:sin_2_hour"] <- x * s2
  g["ndvi:cos_2_hour"] <- x * c2

  # quadratic term:  $x^2 * (_{ndvi\_sq} + \text{harmonics})$ 
  g["ndvi_sq"] <- x^2
  g["ndvi_sq:sin_1_hour"] <- x^2 * s1
  g["ndvi_sq:cos_1_hour"] <- x^2 * c1
  g["ndvi_sq:sin_2_hour"] <- x^2 * s2
  g["ndvi_sq:cos_2_hour"] <- x^2 * c2

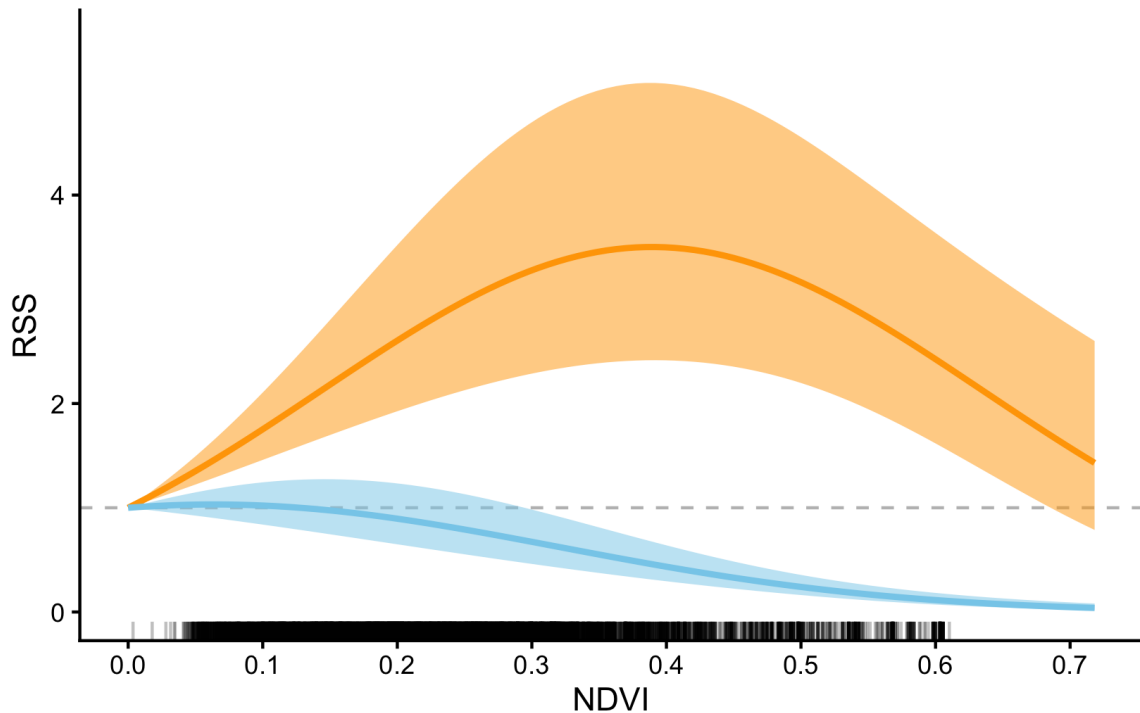
  log_rss <- as.numeric(g %*% betas)
  se_log <- sqrt(as.numeric(t(g) %*% V %*% g))

  data.frame(
    hour = sample_hour,
    ndvi = x,
    log_rss = log_rss,
    rss = exp(log_rss),
    # delta method on the RSS scale: SE_rss = exp(log_rss) * SE_log
    rss_lwr = exp(log_rss - 1.96 * se_log),
    rss_upr = exp(log_rss + 1.96 * se_log)
  )
})

# plot the response curve
ggplot(curve_df, aes(x = ndvi)) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(ymin = rss_lwr, ymax = rss_upr, fill = as.factor(hour)),
    alpha = 0.5) +
  geom_line(aes(y = rss, colour = as.factor(hour)), linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +

```

```
scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
scale_y_continuous(limits = c(0, 5.5)) +
labs(x = "NDVI", y = "RSS") +
theme_classic() +
theme(legend.position = "none")
```



```
ggsave(paste0(plot_save_path, "/ndvi_quadratic_response_sample_hour_id", which_buffalo, ".png"),
        width = 70, height = 80, units = "mm", dpi = 600)
```

Temporal dynamics with smooth terms

```
# fit an ssf gam model
# We set the knots explicitly at 0 and 24. Without this, mgcv places the cyclic
# endpoints at the range of the data (0 and 23), which wraps hour 23 onto hour 0
# and squeezes an hour out of the daily cycle.
ssf_gam_temporal <- mgcv::gam(cbind(times, step_id_) ~
    s(hour, by = ndvi, bs = "cc") + #, k = 5
    sl_ + log_sl + cos_ta,
    knots = list(hour = c(0, 24)),
    data = buffalo_ssf_data,
```

```

        family = cox.ph,
        weight = case_)

summary(ssf_gam_temporal)

```

Family: Cox PH

Link function: identity

Formula:

```
cbind(times, step_id_) ~ s(hour, by = ndvi, bs = "cc") + sl_ +
  log_sl + cos_ta
```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-1.179e-05	3.196e-05	-0.369	0.712
log_sl	7.320e-06	5.702e-03	0.001	0.999
cos_ta	-5.380e-03	1.504e-02	-0.358	0.721

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
s(hour):ndvi	8.191	8.819	154.5	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = -3.61%

-REML = 23204 Scale est. = 1 n = 106678

```
AIC(ssf_gam_temporal)
```

```
[1] 46362.27
```

```
gam_smooths <- smooth_estimates(ssf_gam_temporal)
```

```

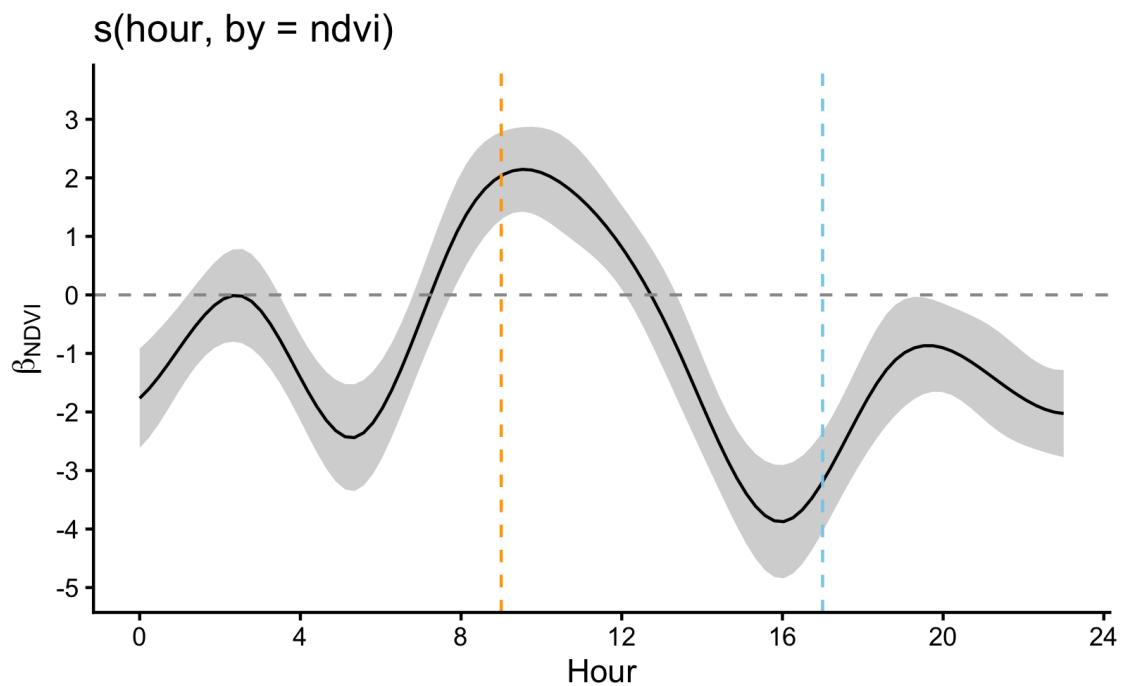
gratia::draw(ssf_gam_temporal, rug = FALSE) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey60") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  labs() +
  scale_x_continuous(breaks = seq(0, 24, by = 4)) +
  scale_y_continuous(limits = c(-5, 3.5), breaks = seq(-5, 4, by = 1)) +
  labs(title = "s(hour, by = ndvi)",

```

```

    subtitle = NULL,
    caption = NULL,
    x = "Hour",
    y = expression(beta[NDVI]), fill = NULL) +
    theme_classic()

```



```

ggsave(paste0(plot_save_path, "/ndvi_gam_coef_response_id", which_buffalo, ".png"),
        width = 150, height = 80, units = "mm", dpi = 600)

```

Pull out a response curve from one of the hours

```

pred_data_hour9 <- pred_data
pred_data_hour9$hour <- 9

pred_data_hour17 <- pred_data
pred_data_hour17$hour <- 17

pred_data_hour9_17 <- bind_rows(pred_data_hour9, pred_data_hour17)

# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam_temporal,

```

```

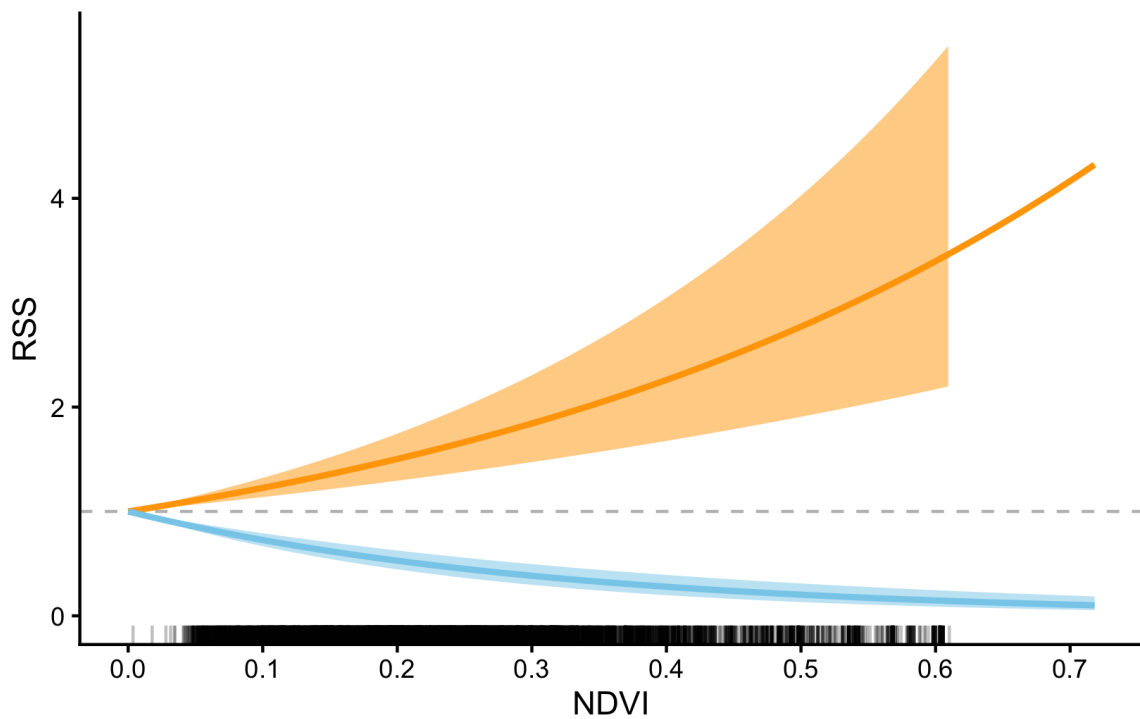
        newdata = pred_data_hour9_17,
        type = "link", # log-scale
        se.fit = TRUE)

pred_data_hour9_17 <- pred_data_hour9_17 %>%
  mutate(
    # Extract log RSS and standard errors
    log_rss = predictions$fit,
    se_log_rss = predictions$se.fit,
    # Calculate confidence intervals on log-scale
    log_rss_lower = log_rss - 1.96 * se_log_rss,
    log_rss_upper = log_rss + 1.96 * se_log_rss,
    # Transform to RSS scale
    rss = exp(log_rss),
    rss_lower = exp(log_rss_lower),
    rss_upper = exp(log_rss_upper)
  )

# plot the response curve
ggplot(data = pred_data_hour9_17) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi,
                 ymin = rss_lower, ymax = rss_upper,
                 fill = as.factor(hour)),
            alpha = 0.5) +
  geom_line(aes(x = ndvi, y = rss, colour = as.factor(hour)),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
           aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```

Warning: Removed 15 rows containing missing values or values outside the scale range (`geom\_ribbon()`).



```
ggsave(paste0(plot_save_path, "/ndvi_gam_response_sample_hour_id", which_buffalo, ".png"),
       width = 70, height = 80, units = "mm", dpi = 600)
```

Warning: Removed 15 rows containing missing values or values outside the scale range (``geom_ribbon()``).

GAM selection surface

```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                               hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  sl_ = 0,
  log_sl = 0,
  cos_ta = 0
)

ndvi_hour_values
```

```
ndvi      hour sl_ log_sl cos_ta
```

1	0.000000000	0.0000000	0	0	0
2	0.007254508	0.0000000	0	0	0
3	0.014509015	0.0000000	0	0	0
4	0.021763523	0.0000000	0	0	0
5	0.029018030	0.0000000	0	0	0
6	0.036272538	0.0000000	0	0	0
7	0.043527045	0.0000000	0	0	0
8	0.050781553	0.0000000	0	0	0
9	0.058036060	0.0000000	0	0	0
10	0.065290568	0.0000000	0	0	0
11	0.072545076	0.0000000	0	0	0
12	0.079799583	0.0000000	0	0	0
13	0.087054091	0.0000000	0	0	0
14	0.094308598	0.0000000	0	0	0
15	0.101563106	0.0000000	0	0	0
16	0.108817613	0.0000000	0	0	0
17	0.116072121	0.0000000	0	0	0
18	0.123326628	0.0000000	0	0	0
19	0.130581136	0.0000000	0	0	0
20	0.137835643	0.0000000	0	0	0
21	0.145090151	0.0000000	0	0	0
22	0.152344659	0.0000000	0	0	0
23	0.159599166	0.0000000	0	0	0
24	0.166853674	0.0000000	0	0	0
25	0.174108181	0.0000000	0	0	0
26	0.181362689	0.0000000	0	0	0
27	0.188617196	0.0000000	0	0	0
28	0.195871704	0.0000000	0	0	0
29	0.203126211	0.0000000	0	0	0
30	0.210380719	0.0000000	0	0	0
31	0.217635227	0.0000000	0	0	0
32	0.224889734	0.0000000	0	0	0
33	0.232144242	0.0000000	0	0	0
34	0.239398749	0.0000000	0	0	0
35	0.246653257	0.0000000	0	0	0
36	0.253907764	0.0000000	0	0	0
37	0.261162272	0.0000000	0	0	0
38	0.268416779	0.0000000	0	0	0
39	0.275671287	0.0000000	0	0	0
40	0.282925794	0.0000000	0	0	0
41	0.290180302	0.0000000	0	0	0
42	0.297434810	0.0000000	0	0	0
43	0.304689317	0.0000000	0	0	0
44	0.311943825	0.0000000	0	0	0
45	0.319198332	0.0000000	0	0	0

46	0.326452840	0.0000000	0	0	0
47	0.333707347	0.0000000	0	0	0
48	0.340961855	0.0000000	0	0	0
49	0.348216362	0.0000000	0	0	0
50	0.355470870	0.0000000	0	0	0
51	0.362725378	0.0000000	0	0	0
52	0.369979885	0.0000000	0	0	0
53	0.377234393	0.0000000	0	0	0
54	0.384488900	0.0000000	0	0	0
55	0.391743408	0.0000000	0	0	0
56	0.398997915	0.0000000	0	0	0
57	0.406252423	0.0000000	0	0	0
58	0.413506930	0.0000000	0	0	0
59	0.420761438	0.0000000	0	0	0
60	0.428015945	0.0000000	0	0	0
61	0.435270453	0.0000000	0	0	0
62	0.442524961	0.0000000	0	0	0
63	0.449779468	0.0000000	0	0	0
64	0.457033976	0.0000000	0	0	0
65	0.464288483	0.0000000	0	0	0
66	0.471542991	0.0000000	0	0	0
67	0.478797498	0.0000000	0	0	0
68	0.486052006	0.0000000	0	0	0
69	0.493306513	0.0000000	0	0	0
70	0.500561021	0.0000000	0	0	0
71	0.507815529	0.0000000	0	0	0
72	0.515070036	0.0000000	0	0	0
73	0.522324544	0.0000000	0	0	0
74	0.529579051	0.0000000	0	0	0
75	0.536833559	0.0000000	0	0	0
76	0.544088066	0.0000000	0	0	0
77	0.551342574	0.0000000	0	0	0
78	0.558597081	0.0000000	0	0	0
79	0.565851589	0.0000000	0	0	0
80	0.573106097	0.0000000	0	0	0
81	0.580360604	0.0000000	0	0	0
82	0.587615112	0.0000000	0	0	0
83	0.594869619	0.0000000	0	0	0
84	0.602124127	0.0000000	0	0	0
85	0.609378634	0.0000000	0	0	0
86	0.616633142	0.0000000	0	0	0
87	0.623887649	0.0000000	0	0	0
88	0.631142157	0.0000000	0	0	0
89	0.638396664	0.0000000	0	0	0
90	0.645651172	0.0000000	0	0	0

91	0.652905680	0.0000000	0	0	0
92	0.660160187	0.0000000	0	0	0
93	0.667414695	0.0000000	0	0	0
94	0.674669202	0.0000000	0	0	0
95	0.681923710	0.0000000	0	0	0
96	0.689178217	0.0000000	0	0	0
97	0.696432725	0.0000000	0	0	0
98	0.703687232	0.0000000	0	0	0
99	0.710941740	0.0000000	0	0	0
100	0.718196248	0.0000000	0	0	0
101	0.000000000	0.2323232	0	0	0
102	0.007254508	0.2323232	0	0	0
103	0.014509015	0.2323232	0	0	0
104	0.021763523	0.2323232	0	0	0
105	0.029018030	0.2323232	0	0	0
106	0.036272538	0.2323232	0	0	0
107	0.043527045	0.2323232	0	0	0
108	0.050781553	0.2323232	0	0	0
109	0.058036060	0.2323232	0	0	0
110	0.065290568	0.2323232	0	0	0
111	0.072545076	0.2323232	0	0	0
112	0.079799583	0.2323232	0	0	0
113	0.087054091	0.2323232	0	0	0
114	0.094308598	0.2323232	0	0	0
115	0.101563106	0.2323232	0	0	0
116	0.108817613	0.2323232	0	0	0
117	0.116072121	0.2323232	0	0	0
118	0.123326628	0.2323232	0	0	0
119	0.130581136	0.2323232	0	0	0
120	0.137835643	0.2323232	0	0	0
121	0.145090151	0.2323232	0	0	0
122	0.152344659	0.2323232	0	0	0
123	0.159599166	0.2323232	0	0	0
124	0.166853674	0.2323232	0	0	0
125	0.174108181	0.2323232	0	0	0
126	0.181362689	0.2323232	0	0	0
127	0.188617196	0.2323232	0	0	0
128	0.195871704	0.2323232	0	0	0
129	0.203126211	0.2323232	0	0	0
130	0.210380719	0.2323232	0	0	0
131	0.217635227	0.2323232	0	0	0
132	0.224889734	0.2323232	0	0	0
133	0.232144242	0.2323232	0	0	0
134	0.239398749	0.2323232	0	0	0
135	0.246653257	0.2323232	0	0	0

136	0.253907764	0.2323232	0	0	0
137	0.261162272	0.2323232	0	0	0
138	0.268416779	0.2323232	0	0	0
139	0.275671287	0.2323232	0	0	0
140	0.282925794	0.2323232	0	0	0
141	0.290180302	0.2323232	0	0	0
142	0.297434810	0.2323232	0	0	0
143	0.304689317	0.2323232	0	0	0
144	0.311943825	0.2323232	0	0	0
145	0.319198332	0.2323232	0	0	0
146	0.326452840	0.2323232	0	0	0
147	0.333707347	0.2323232	0	0	0
148	0.340961855	0.2323232	0	0	0
149	0.348216362	0.2323232	0	0	0
150	0.355470870	0.2323232	0	0	0
151	0.362725378	0.2323232	0	0	0
152	0.369979885	0.2323232	0	0	0
153	0.377234393	0.2323232	0	0	0
154	0.384488900	0.2323232	0	0	0
155	0.391743408	0.2323232	0	0	0
156	0.398997915	0.2323232	0	0	0
157	0.406252423	0.2323232	0	0	0
158	0.413506930	0.2323232	0	0	0
159	0.420761438	0.2323232	0	0	0
160	0.428015945	0.2323232	0	0	0
161	0.435270453	0.2323232	0	0	0
162	0.442524961	0.2323232	0	0	0
163	0.449779468	0.2323232	0	0	0
164	0.457033976	0.2323232	0	0	0
165	0.464288483	0.2323232	0	0	0
166	0.471542991	0.2323232	0	0	0
167	0.478797498	0.2323232	0	0	0
168	0.486052006	0.2323232	0	0	0
169	0.493306513	0.2323232	0	0	0
170	0.500561021	0.2323232	0	0	0
171	0.507815529	0.2323232	0	0	0
172	0.515070036	0.2323232	0	0	0
173	0.522324544	0.2323232	0	0	0
174	0.529579051	0.2323232	0	0	0
175	0.536833559	0.2323232	0	0	0
176	0.544088066	0.2323232	0	0	0
177	0.551342574	0.2323232	0	0	0
178	0.558597081	0.2323232	0	0	0
179	0.565851589	0.2323232	0	0	0
180	0.573106097	0.2323232	0	0	0

181	0.580360604	0.2323232	0	0	0
182	0.587615112	0.2323232	0	0	0
183	0.594869619	0.2323232	0	0	0
184	0.602124127	0.2323232	0	0	0
185	0.609378634	0.2323232	0	0	0
186	0.616633142	0.2323232	0	0	0
187	0.623887649	0.2323232	0	0	0
188	0.631142157	0.2323232	0	0	0
189	0.638396664	0.2323232	0	0	0
190	0.645651172	0.2323232	0	0	0
191	0.652905680	0.2323232	0	0	0
192	0.660160187	0.2323232	0	0	0
193	0.667414695	0.2323232	0	0	0
194	0.674669202	0.2323232	0	0	0
195	0.681923710	0.2323232	0	0	0
196	0.689178217	0.2323232	0	0	0
197	0.696432725	0.2323232	0	0	0
198	0.703687232	0.2323232	0	0	0
199	0.710941740	0.2323232	0	0	0
200	0.718196248	0.2323232	0	0	0
201	0.000000000	0.4646465	0	0	0
202	0.007254508	0.4646465	0	0	0
203	0.014509015	0.4646465	0	0	0
204	0.021763523	0.4646465	0	0	0
205	0.029018030	0.4646465	0	0	0
206	0.036272538	0.4646465	0	0	0
207	0.043527045	0.4646465	0	0	0
208	0.050781553	0.4646465	0	0	0
209	0.058036060	0.4646465	0	0	0
210	0.065290568	0.4646465	0	0	0
211	0.072545076	0.4646465	0	0	0
212	0.079799583	0.4646465	0	0	0
213	0.087054091	0.4646465	0	0	0
214	0.094308598	0.4646465	0	0	0
215	0.101563106	0.4646465	0	0	0
216	0.108817613	0.4646465	0	0	0
217	0.116072121	0.4646465	0	0	0
218	0.123326628	0.4646465	0	0	0
219	0.130581136	0.4646465	0	0	0
220	0.137835643	0.4646465	0	0	0
221	0.145090151	0.4646465	0	0	0
222	0.152344659	0.4646465	0	0	0
223	0.159599166	0.4646465	0	0	0
224	0.166853674	0.4646465	0	0	0
225	0.174108181	0.4646465	0	0	0

226	0.181362689	0.4646465	0	0	0
227	0.188617196	0.4646465	0	0	0
228	0.195871704	0.4646465	0	0	0
229	0.203126211	0.4646465	0	0	0
230	0.210380719	0.4646465	0	0	0
231	0.217635227	0.4646465	0	0	0
232	0.224889734	0.4646465	0	0	0
233	0.232144242	0.4646465	0	0	0
234	0.239398749	0.4646465	0	0	0
235	0.246653257	0.4646465	0	0	0
236	0.253907764	0.4646465	0	0	0
237	0.261162272	0.4646465	0	0	0
238	0.268416779	0.4646465	0	0	0
239	0.275671287	0.4646465	0	0	0
240	0.282925794	0.4646465	0	0	0
241	0.290180302	0.4646465	0	0	0
242	0.297434810	0.4646465	0	0	0
243	0.304689317	0.4646465	0	0	0
244	0.311943825	0.4646465	0	0	0
245	0.319198332	0.4646465	0	0	0
246	0.326452840	0.4646465	0	0	0
247	0.333707347	0.4646465	0	0	0
248	0.340961855	0.4646465	0	0	0
249	0.348216362	0.4646465	0	0	0
250	0.355470870	0.4646465	0	0	0
251	0.362725378	0.4646465	0	0	0
252	0.369979885	0.4646465	0	0	0
253	0.377234393	0.4646465	0	0	0
254	0.384488900	0.4646465	0	0	0
255	0.391743408	0.4646465	0	0	0
256	0.398997915	0.4646465	0	0	0
257	0.406252423	0.4646465	0	0	0
258	0.413506930	0.4646465	0	0	0
259	0.420761438	0.4646465	0	0	0
260	0.428015945	0.4646465	0	0	0
261	0.435270453	0.4646465	0	0	0
262	0.442524961	0.4646465	0	0	0
263	0.449779468	0.4646465	0	0	0
264	0.457033976	0.4646465	0	0	0
265	0.464288483	0.4646465	0	0	0
266	0.471542991	0.4646465	0	0	0
267	0.478797498	0.4646465	0	0	0
268	0.486052006	0.4646465	0	0	0
269	0.493306513	0.4646465	0	0	0
270	0.500561021	0.4646465	0	0	0

271	0.507815529	0.4646465	0	0	0
272	0.515070036	0.4646465	0	0	0
273	0.522324544	0.4646465	0	0	0
274	0.529579051	0.4646465	0	0	0
275	0.536833559	0.4646465	0	0	0
276	0.544088066	0.4646465	0	0	0
277	0.551342574	0.4646465	0	0	0
278	0.558597081	0.4646465	0	0	0
279	0.565851589	0.4646465	0	0	0
280	0.573106097	0.4646465	0	0	0
281	0.580360604	0.4646465	0	0	0
282	0.587615112	0.4646465	0	0	0
283	0.594869619	0.4646465	0	0	0
284	0.602124127	0.4646465	0	0	0
285	0.609378634	0.4646465	0	0	0
286	0.616633142	0.4646465	0	0	0
287	0.623887649	0.4646465	0	0	0
288	0.631142157	0.4646465	0	0	0
289	0.638396664	0.4646465	0	0	0
290	0.645651172	0.4646465	0	0	0
291	0.652905680	0.4646465	0	0	0
292	0.660160187	0.4646465	0	0	0
293	0.667414695	0.4646465	0	0	0
294	0.674669202	0.4646465	0	0	0
295	0.681923710	0.4646465	0	0	0
296	0.689178217	0.4646465	0	0	0
297	0.696432725	0.4646465	0	0	0
298	0.703687232	0.4646465	0	0	0
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306	0.036272538	0.6969697	0	0	0
307	0.043527045	0.6969697	0	0	0
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309	0.058036060	0.6969697	0	0	0
310	0.065290568	0.6969697	0	0	0
311	0.072545076	0.6969697	0	0	0
312	0.079799583	0.6969697	0	0	0
313	0.087054091	0.6969697	0	0	0
314	0.094308598	0.6969697	0	0	0
315	0.101563106	0.6969697	0	0	0

316	0.108817613	0.6969697	0	0	0
317	0.116072121	0.6969697	0	0	0
318	0.123326628	0.6969697	0	0	0
319	0.130581136	0.6969697	0	0	0
320	0.137835643	0.6969697	0	0	0
321	0.145090151	0.6969697	0	0	0
322	0.152344659	0.6969697	0	0	0
323	0.159599166	0.6969697	0	0	0
324	0.166853674	0.6969697	0	0	0
325	0.174108181	0.6969697	0	0	0
326	0.181362689	0.6969697	0	0	0
327	0.188617196	0.6969697	0	0	0
328	0.195871704	0.6969697	0	0	0
329	0.203126211	0.6969697	0	0	0
330	0.210380719	0.6969697	0	0	0
331	0.217635227	0.6969697	0	0	0
332	0.224889734	0.6969697	0	0	0
333	0.232144242	0.6969697	0	0	0
334	0.239398749	0.6969697	0	0	0
335	0.246653257	0.6969697	0	0	0
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337	0.261162272	0.6969697	0	0	0
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345	0.319198332	0.6969697	0	0	0
346	0.326452840	0.6969697	0	0	0
347	0.333707347	0.6969697	0	0	0
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360	0.428015945	0.6969697	0	0	0

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394	0.674669202	0.6969697	0	0	0
395	0.681923710	0.6969697	0	0	0
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405	0.029018030	0.9292929	0	0	0

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415	0.101563106	0.9292929	0	0	0
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425	0.174108181	0.9292929	0	0	0
426	0.181362689	0.9292929	0	0	0
427	0.188617196	0.9292929	0	0	0
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433	0.232144242	0.9292929	0	0	0
434	0.239398749	0.9292929	0	0	0
435	0.246653257	0.9292929	0	0	0
436	0.253907764	0.9292929	0	0	0
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440	0.282925794	0.9292929	0	0	0
441	0.290180302	0.9292929	0	0	0
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443	0.304689317	0.9292929	0	0	0
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445	0.319198332	0.9292929	0	0	0
446	0.326452840	0.9292929	0	0	0
447	0.333707347	0.9292929	0	0	0
448	0.340961855	0.9292929	0	0	0
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461	0.435270453	0.9292929	0	0	0
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484	0.602124127	0.9292929	0	0	0
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486	0.616633142	0.9292929	0	0	0
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490	0.645651172	0.9292929	0	0	0
491	0.652905680	0.9292929	0	0	0
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495	0.681923710	0.9292929	0	0	0

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615	0.101563106	1.3939394	0	0	0
616	0.108817613	1.3939394	0	0	0
617	0.116072121	1.3939394	0	0	0
618	0.123326628	1.3939394	0	0	0
619	0.130581136	1.3939394	0	0	0
620	0.137835643	1.3939394	0	0	0
621	0.145090151	1.3939394	0	0	0
622	0.152344659	1.3939394	0	0	0
623	0.159599166	1.3939394	0	0	0
624	0.166853674	1.3939394	0	0	0
625	0.174108181	1.3939394	0	0	0
626	0.181362689	1.3939394	0	0	0
627	0.188617196	1.3939394	0	0	0
628	0.195871704	1.3939394	0	0	0
629	0.203126211	1.3939394	0	0	0
630	0.210380719	1.3939394	0	0	0

631	0.217635227	1.3939394	0	0	0
632	0.224889734	1.3939394	0	0	0
633	0.232144242	1.3939394	0	0	0
634	0.239398749	1.3939394	0	0	0
635	0.246653257	1.3939394	0	0	0
636	0.253907764	1.3939394	0	0	0
637	0.261162272	1.3939394	0	0	0
638	0.268416779	1.3939394	0	0	0
639	0.275671287	1.3939394	0	0	0
640	0.282925794	1.3939394	0	0	0
641	0.290180302	1.3939394	0	0	0
642	0.297434810	1.3939394	0	0	0
643	0.304689317	1.3939394	0	0	0
644	0.311943825	1.3939394	0	0	0
645	0.319198332	1.3939394	0	0	0
646	0.326452840	1.3939394	0	0	0
647	0.333707347	1.3939394	0	0	0
648	0.340961855	1.3939394	0	0	0
649	0.348216362	1.3939394	0	0	0
650	0.355470870	1.3939394	0	0	0
651	0.362725378	1.3939394	0	0	0
652	0.369979885	1.3939394	0	0	0
653	0.377234393	1.3939394	0	0	0
654	0.384488900	1.3939394	0	0	0
655	0.391743408	1.3939394	0	0	0
656	0.398997915	1.3939394	0	0	0
657	0.406252423	1.3939394	0	0	0
658	0.413506930	1.3939394	0	0	0
659	0.420761438	1.3939394	0	0	0
660	0.428015945	1.3939394	0	0	0
661	0.435270453	1.3939394	0	0	0
662	0.442524961	1.3939394	0	0	0
663	0.449779468	1.3939394	0	0	0
664	0.457033976	1.3939394	0	0	0
665	0.464288483	1.3939394	0	0	0
666	0.471542991	1.3939394	0	0	0
667	0.478797498	1.3939394	0	0	0
668	0.486052006	1.3939394	0	0	0
669	0.493306513	1.3939394	0	0	0
670	0.500561021	1.3939394	0	0	0
671	0.507815529	1.3939394	0	0	0
672	0.515070036	1.3939394	0	0	0
673	0.522324544	1.3939394	0	0	0
674	0.529579051	1.3939394	0	0	0
675	0.536833559	1.3939394	0	0	0

676	0.544088066	1.3939394	0	0	0
677	0.551342574	1.3939394	0	0	0
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679	0.565851589	1.3939394	0	0	0
680	0.573106097	1.3939394	0	0	0
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683	0.594869619	1.3939394	0	0	0
684	0.602124127	1.3939394	0	0	0
685	0.609378634	1.3939394	0	0	0
686	0.616633142	1.3939394	0	0	0
687	0.623887649	1.3939394	0	0	0
688	0.631142157	1.3939394	0	0	0
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695	0.681923710	1.3939394	0	0	0
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718	0.123326628	1.6262626	0	0	0
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724	0.166853674	1.6262626	0	0	0
725	0.174108181	1.6262626	0	0	0
726	0.181362689	1.6262626	0	0	0
727	0.188617196	1.6262626	0	0	0
728	0.195871704	1.6262626	0	0	0
729	0.203126211	1.6262626	0	0	0
730	0.210380719	1.6262626	0	0	0
731	0.217635227	1.6262626	0	0	0
732	0.224889734	1.6262626	0	0	0
733	0.232144242	1.6262626	0	0	0
734	0.239398749	1.6262626	0	0	0
735	0.246653257	1.6262626	0	0	0
736	0.253907764	1.6262626	0	0	0
737	0.261162272	1.6262626	0	0	0
738	0.268416779	1.6262626	0	0	0
739	0.275671287	1.6262626	0	0	0
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741	0.290180302	1.6262626	0	0	0
742	0.297434810	1.6262626	0	0	0
743	0.304689317	1.6262626	0	0	0
744	0.311943825	1.6262626	0	0	0
745	0.319198332	1.6262626	0	0	0
746	0.326452840	1.6262626	0	0	0
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748	0.340961855	1.6262626	0	0	0
749	0.348216362	1.6262626	0	0	0
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751	0.362725378	1.6262626	0	0	0
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754	0.384488900	1.6262626	0	0	0
755	0.391743408	1.6262626	0	0	0
756	0.398997915	1.6262626	0	0	0
757	0.406252423	1.6262626	0	0	0
758	0.413506930	1.6262626	0	0	0
759	0.420761438	1.6262626	0	0	0
760	0.428015945	1.6262626	0	0	0
761	0.435270453	1.6262626	0	0	0
762	0.442524961	1.6262626	0	0	0
763	0.449779468	1.6262626	0	0	0
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795	0.681923710	1.6262626	0	0	0
796	0.689178217	1.6262626	0	0	0
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818	0.123326628	1.8585859	0	0	0
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827	0.188617196	1.8585859	0	0	0
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844	0.311943825	1.8585859	0	0	0
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862	0.442524961	1.8585859	0	0	0
863	0.449779468	1.8585859	0	0	0
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865	0.464288483	1.8585859	0	0	0
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884	0.602124127	1.8585859	0	0	0
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887	0.623887649	1.8585859	0	0	0
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926	0.181362689	2.0909091	0	0	0
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945	0.319198332	2.0909091	0	0	0

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954	0.384488900	2.0909091	0	0	0
955	0.391743408	2.0909091	0	0	0
956	0.398997915	2.0909091	0	0	0
957	0.406252423	2.0909091	0	0	0
958	0.413506930	2.0909091	0	0	0
959	0.420761438	2.0909091	0	0	0
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961	0.435270453	2.0909091	0	0	0
962	0.442524961	2.0909091	0	0	0
963	0.449779468	2.0909091	0	0	0
964	0.457033976	2.0909091	0	0	0
965	0.464288483	2.0909091	0	0	0
966	0.471542991	2.0909091	0	0	0
967	0.478797498	2.0909091	0	0	0
968	0.486052006	2.0909091	0	0	0
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976	0.544088066	2.0909091	0	0	0
977	0.551342574	2.0909091	0	0	0
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979	0.565851589	2.0909091	0	0	0
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984	0.602124127	2.0909091	0	0	0
985	0.609378634	2.0909091	0	0	0
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989	0.638396664	2.0909091	0	0	0
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994	0.674669202	2.0909091	0	0	0
995	0.681923710	2.0909091	0	0	0
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1349	0.348216362	3.0202020	0	0	0
1350	0.355470870	3.0202020	0	0	0

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1352	0.369979885	3.0202020	0	0	0
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1354	0.384488900	3.0202020	0	0	0
1355	0.391743408	3.0202020	0	0	0
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1357	0.406252423	3.0202020	0	0	0
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1710	0.065290568	3.9494949	0	0	0

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1713	0.087054091	3.9494949	0	0	0
1714	0.094308598	3.9494949	0	0	0
1715	0.101563106	3.9494949	0	0	0
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1717	0.116072121	3.9494949	0	0	0
1718	0.123326628	3.9494949	0	0	0
1719	0.130581136	3.9494949	0	0	0
1720	0.137835643	3.9494949	0	0	0
1721	0.145090151	3.9494949	0	0	0
1722	0.152344659	3.9494949	0	0	0
1723	0.159599166	3.9494949	0	0	0
1724	0.166853674	3.9494949	0	0	0
1725	0.174108181	3.9494949	0	0	0
1726	0.181362689	3.9494949	0	0	0
1727	0.188617196	3.9494949	0	0	0
1728	0.195871704	3.9494949	0	0	0
1729	0.203126211	3.9494949	0	0	0
1730	0.210380719	3.9494949	0	0	0
1731	0.217635227	3.9494949	0	0	0
1732	0.224889734	3.9494949	0	0	0
1733	0.232144242	3.9494949	0	0	0
1734	0.239398749	3.9494949	0	0	0
1735	0.246653257	3.9494949	0	0	0
1736	0.253907764	3.9494949	0	0	0
1737	0.261162272	3.9494949	0	0	0
1738	0.268416779	3.9494949	0	0	0
1739	0.275671287	3.9494949	0	0	0
1740	0.282925794	3.9494949	0	0	0
1741	0.290180302	3.9494949	0	0	0
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1743	0.304689317	3.9494949	0	0	0
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1745	0.319198332	3.9494949	0	0	0
1746	0.326452840	3.9494949	0	0	0
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1806	0.036272538	4.1818182	0	0	0
1807	0.043527045	4.1818182	0	0	0
1808	0.050781553	4.1818182	0	0	0
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1810	0.065290568	4.1818182	0	0	0
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1812	0.079799583	4.1818182	0	0	0
1813	0.087054091	4.1818182	0	0	0
1814	0.094308598	4.1818182	0	0	0
1815	0.101563106	4.1818182	0	0	0
1816	0.108817613	4.1818182	0	0	0
1817	0.116072121	4.1818182	0	0	0
1818	0.123326628	4.1818182	0	0	0
1819	0.130581136	4.1818182	0	0	0
1820	0.137835643	4.1818182	0	0	0
1821	0.145090151	4.1818182	0	0	0
1822	0.152344659	4.1818182	0	0	0
1823	0.159599166	4.1818182	0	0	0
1824	0.166853674	4.1818182	0	0	0
1825	0.174108181	4.1818182	0	0	0
1826	0.181362689	4.1818182	0	0	0
1827	0.188617196	4.1818182	0	0	0
1828	0.195871704	4.1818182	0	0	0
1829	0.203126211	4.1818182	0	0	0
1830	0.210380719	4.1818182	0	0	0
1831	0.217635227	4.1818182	0	0	0
1832	0.224889734	4.1818182	0	0	0
1833	0.232144242	4.1818182	0	0	0
1834	0.239398749	4.1818182	0	0	0
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1836	0.253907764	4.1818182	0	0	0
1837	0.261162272	4.1818182	0	0	0
1838	0.268416779	4.1818182	0	0	0
1839	0.275671287	4.1818182	0	0	0
1840	0.282925794	4.1818182	0	0	0
1841	0.290180302	4.1818182	0	0	0
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1843	0.304689317	4.1818182	0	0	0
1844	0.311943825	4.1818182	0	0	0
1845	0.319198332	4.1818182	0	0	0

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1848	0.340961855	4.1818182	0	0	0
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1915	0.101563106	4.4141414	0	0	0
1916	0.108817613	4.4141414	0	0	0
1917	0.116072121	4.4141414	0	0	0
1918	0.123326628	4.4141414	0	0	0
1919	0.130581136	4.4141414	0	0	0
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1921	0.145090151	4.4141414	0	0	0
1922	0.152344659	4.4141414	0	0	0
1923	0.159599166	4.4141414	0	0	0
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1926	0.181362689	4.4141414	0	0	0
1927	0.188617196	4.4141414	0	0	0
1928	0.195871704	4.4141414	0	0	0
1929	0.203126211	4.4141414	0	0	0
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1931	0.217635227	4.4141414	0	0	0
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1933	0.232144242	4.4141414	0	0	0
1934	0.239398749	4.4141414	0	0	0
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1938	0.268416779	4.4141414	0	0	0
1939	0.275671287	4.4141414	0	0	0
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1941	0.290180302	4.4141414	0	0	0
1942	0.297434810	4.4141414	0	0	0
1943	0.304689317	4.4141414	0	0	0
1944	0.311943825	4.4141414	0	0	0
1945	0.319198332	4.4141414	0	0	0
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1950	0.355470870	4.4141414	0	0	0
1951	0.362725378	4.4141414	0	0	0
1952	0.369979885	4.4141414	0	0	0
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1954	0.384488900	4.4141414	0	0	0
1955	0.391743408	4.4141414	0	0	0
1956	0.398997915	4.4141414	0	0	0
1957	0.406252423	4.4141414	0	0	0
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1959	0.420761438	4.4141414	0	0	0
1960	0.428015945	4.4141414	0	0	0
1961	0.435270453	4.4141414	0	0	0
1962	0.442524961	4.4141414	0	0	0
1963	0.449779468	4.4141414	0	0	0
1964	0.457033976	4.4141414	0	0	0
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1966	0.471542991	4.4141414	0	0	0
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1985	0.609378634	4.4141414	0	0	0
1986	0.616633142	4.4141414	0	0	0
1987	0.623887649	4.4141414	0	0	0
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1991	0.652905680	4.4141414	0	0	0
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1993	0.667414695	4.4141414	0	0	0
1994	0.674669202	4.4141414	0	0	0
1995	0.681923710	4.4141414	0	0	0
1996	0.689178217	4.4141414	0	0	0
1997	0.696432725	4.4141414	0	0	0
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2005	0.029018030	4.6464646	0	0	0
2006	0.036272538	4.6464646	0	0	0
2007	0.043527045	4.6464646	0	0	0
2008	0.050781553	4.6464646	0	0	0
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2010	0.065290568	4.6464646	0	0	0
2011	0.072545076	4.6464646	0	0	0
2012	0.079799583	4.6464646	0	0	0
2013	0.087054091	4.6464646	0	0	0
2014	0.094308598	4.6464646	0	0	0
2015	0.101563106	4.6464646	0	0	0
2016	0.108817613	4.6464646	0	0	0
2017	0.116072121	4.6464646	0	0	0
2018	0.123326628	4.6464646	0	0	0
2019	0.130581136	4.6464646	0	0	0
2020	0.137835643	4.6464646	0	0	0
2021	0.145090151	4.6464646	0	0	0
2022	0.152344659	4.6464646	0	0	0
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2025	0.174108181	4.6464646	0	0	0

2026	0.181362689	4.6464646	0	0	0
2027	0.188617196	4.6464646	0	0	0
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2039	0.275671287	4.6464646	0	0	0
2040	0.282925794	4.6464646	0	0	0
2041	0.290180302	4.6464646	0	0	0
2042	0.297434810	4.6464646	0	0	0
2043	0.304689317	4.6464646	0	0	0
2044	0.311943825	4.6464646	0	0	0
2045	0.319198332	4.6464646	0	0	0
2046	0.326452840	4.6464646	0	0	0
2047	0.333707347	4.6464646	0	0	0
2048	0.340961855	4.6464646	0	0	0
2049	0.348216362	4.6464646	0	0	0
2050	0.355470870	4.6464646	0	0	0
2051	0.362725378	4.6464646	0	0	0
2052	0.369979885	4.6464646	0	0	0
2053	0.377234393	4.6464646	0	0	0
2054	0.384488900	4.6464646	0	0	0
2055	0.391743408	4.6464646	0	0	0
2056	0.398997915	4.6464646	0	0	0
2057	0.406252423	4.6464646	0	0	0
2058	0.413506930	4.6464646	0	0	0
2059	0.420761438	4.6464646	0	0	0
2060	0.428015945	4.6464646	0	0	0
2061	0.435270453	4.6464646	0	0	0
2062	0.442524961	4.6464646	0	0	0
2063	0.449779468	4.6464646	0	0	0
2064	0.457033976	4.6464646	0	0	0
2065	0.464288483	4.6464646	0	0	0
2066	0.471542991	4.6464646	0	0	0
2067	0.478797498	4.6464646	0	0	0
2068	0.486052006	4.6464646	0	0	0
2069	0.493306513	4.6464646	0	0	0
2070	0.500561021	4.6464646	0	0	0

2071	0.507815529	4.6464646	0	0	0
2072	0.515070036	4.6464646	0	0	0
2073	0.522324544	4.6464646	0	0	0
2074	0.529579051	4.6464646	0	0	0
2075	0.536833559	4.6464646	0	0	0
2076	0.544088066	4.6464646	0	0	0
2077	0.551342574	4.6464646	0	0	0
2078	0.558597081	4.6464646	0	0	0
2079	0.565851589	4.6464646	0	0	0
2080	0.573106097	4.6464646	0	0	0
2081	0.580360604	4.6464646	0	0	0
2082	0.587615112	4.6464646	0	0	0
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2084	0.602124127	4.6464646	0	0	0
2085	0.609378634	4.6464646	0	0	0
2086	0.616633142	4.6464646	0	0	0
2087	0.623887649	4.6464646	0	0	0
2088	0.631142157	4.6464646	0	0	0
2089	0.638396664	4.6464646	0	0	0
2090	0.645651172	4.6464646	0	0	0
2091	0.652905680	4.6464646	0	0	0
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2094	0.674669202	4.6464646	0	0	0
2095	0.681923710	4.6464646	0	0	0
2096	0.689178217	4.6464646	0	0	0
2097	0.696432725	4.6464646	0	0	0
2098	0.703687232	4.6464646	0	0	0
2099	0.710941740	4.6464646	0	0	0
2100	0.718196248	4.6464646	0	0	0
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2102	0.007254508	4.8787879	0	0	0
2103	0.014509015	4.8787879	0	0	0
2104	0.021763523	4.8787879	0	0	0
2105	0.029018030	4.8787879	0	0	0
2106	0.036272538	4.8787879	0	0	0
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2110	0.065290568	4.8787879	0	0	0
2111	0.072545076	4.8787879	0	0	0
2112	0.079799583	4.8787879	0	0	0
2113	0.087054091	4.8787879	0	0	0
2114	0.094308598	4.8787879	0	0	0
2115	0.101563106	4.8787879	0	0	0

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2119	0.130581136	4.8787879	0	0	0
2120	0.137835643	4.8787879	0	0	0
2121	0.145090151	4.8787879	0	0	0
2122	0.152344659	4.8787879	0	0	0
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2125	0.174108181	4.8787879	0	0	0
2126	0.181362689	4.8787879	0	0	0
2127	0.188617196	4.8787879	0	0	0
2128	0.195871704	4.8787879	0	0	0
2129	0.203126211	4.8787879	0	0	0
2130	0.210380719	4.8787879	0	0	0
2131	0.217635227	4.8787879	0	0	0
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2133	0.232144242	4.8787879	0	0	0
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2136	0.253907764	4.8787879	0	0	0
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2138	0.268416779	4.8787879	0	0	0
2139	0.275671287	4.8787879	0	0	0
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2159	0.420761438	4.8787879	0	0	0
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2165	0.464288483	4.8787879	0	0	0
2166	0.471542991	4.8787879	0	0	0
2167	0.478797498	4.8787879	0	0	0
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2169	0.493306513	4.8787879	0	0	0
2170	0.500561021	4.8787879	0	0	0
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2177	0.551342574	4.8787879	0	0	0
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2179	0.565851589	4.8787879	0	0	0
2180	0.573106097	4.8787879	0	0	0
2181	0.580360604	4.8787879	0	0	0
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2185	0.609378634	4.8787879	0	0	0
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2187	0.623887649	4.8787879	0	0	0
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2190	0.645651172	4.8787879	0	0	0
2191	0.652905680	4.8787879	0	0	0
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2193	0.667414695	4.8787879	0	0	0
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2196	0.689178217	4.8787879	0	0	0
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2217	0.116072121	5.1111111	0	0	0
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2219	0.130581136	5.1111111	0	0	0
2220	0.137835643	5.1111111	0	0	0
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2226	0.181362689	5.1111111	0	0	0
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2236	0.253907764	5.1111111	0	0	0
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2385	0.609378634	5.3434343	0	0	0

2386	0.616633142	5.3434343	0	0	0
2387	0.623887649	5.3434343	0	0	0
2388	0.631142157	5.3434343	0	0	0
2389	0.638396664	5.3434343	0	0	0
2390	0.645651172	5.3434343	0	0	0
2391	0.652905680	5.3434343	0	0	0
2392	0.660160187	5.3434343	0	0	0
2393	0.667414695	5.3434343	0	0	0
2394	0.674669202	5.3434343	0	0	0
2395	0.681923710	5.3434343	0	0	0
2396	0.689178217	5.3434343	0	0	0
2397	0.696432725	5.3434343	0	0	0
2398	0.703687232	5.3434343	0	0	0
2399	0.710941740	5.3434343	0	0	0
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2403	0.014509015	5.5757576	0	0	0
2404	0.021763523	5.5757576	0	0	0
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2406	0.036272538	5.5757576	0	0	0
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2409	0.058036060	5.5757576	0	0	0
2410	0.065290568	5.5757576	0	0	0
2411	0.072545076	5.5757576	0	0	0
2412	0.079799583	5.5757576	0	0	0
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2414	0.094308598	5.5757576	0	0	0
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2421	0.145090151	5.5757576	0	0	0
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2785	0.609378634	6.2727273	0	0	0
2786	0.616633142	6.2727273	0	0	0
2787	0.623887649	6.2727273	0	0	0
2788	0.631142157	6.2727273	0	0	0
2789	0.638396664	6.2727273	0	0	0
2790	0.645651172	6.2727273	0	0	0

2791	0.652905680	6.2727273	0	0	0
2792	0.660160187	6.2727273	0	0	0
2793	0.667414695	6.2727273	0	0	0
2794	0.674669202	6.2727273	0	0	0
2795	0.681923710	6.2727273	0	0	0
2796	0.689178217	6.2727273	0	0	0
2797	0.696432725	6.2727273	0	0	0
2798	0.703687232	6.2727273	0	0	0
2799	0.710941740	6.2727273	0	0	0
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2808	0.050781553	6.5050505	0	0	0
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2810	0.065290568	6.5050505	0	0	0
2811	0.072545076	6.5050505	0	0	0
2812	0.079799583	6.5050505	0	0	0
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2814	0.094308598	6.5050505	0	0	0
2815	0.101563106	6.5050505	0	0	0
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2817	0.116072121	6.5050505	0	0	0
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2821	0.145090151	6.5050505	0	0	0
2822	0.152344659	6.5050505	0	0	0
2823	0.159599166	6.5050505	0	0	0
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2825	0.174108181	6.5050505	0	0	0
2826	0.181362689	6.5050505	0	0	0
2827	0.188617196	6.5050505	0	0	0
2828	0.195871704	6.5050505	0	0	0
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2832	0.224889734	6.5050505	0	0	0
2833	0.232144242	6.5050505	0	0	0
2834	0.239398749	6.5050505	0	0	0
2835	0.246653257	6.5050505	0	0	0

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2898	0.703687232	6.5050505	0	0	0
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3118	0.123326628	7.2020202	0	0	0
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3123	0.159599166	7.2020202	0	0	0
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3128	0.195871704	7.2020202	0	0	0
3129	0.203126211	7.2020202	0	0	0
3130	0.210380719	7.2020202	0	0	0
3131	0.217635227	7.2020202	0	0	0
3132	0.224889734	7.2020202	0	0	0
3133	0.232144242	7.2020202	0	0	0
3134	0.239398749	7.2020202	0	0	0
3135	0.246653257	7.2020202	0	0	0
3136	0.253907764	7.2020202	0	0	0
3137	0.261162272	7.2020202	0	0	0
3138	0.268416779	7.2020202	0	0	0
3139	0.275671287	7.2020202	0	0	0
3140	0.282925794	7.2020202	0	0	0
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3856	0.398997915	8.8282828	0	0	0
3857	0.406252423	8.8282828	0	0	0
3858	0.413506930	8.8282828	0	0	0
3859	0.420761438	8.8282828	0	0	0
3860	0.428015945	8.8282828	0	0	0
3861	0.435270453	8.8282828	0	0	0
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3863	0.449779468	8.8282828	0	0	0
3864	0.457033976	8.8282828	0	0	0
3865	0.464288483	8.8282828	0	0	0
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3875	0.536833559	8.8282828	0	0	0
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3880	0.573106097	8.8282828	0	0	0
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3887	0.623887649	8.8282828	0	0	0
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3889	0.638396664	8.8282828	0	0	0
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3893	0.667414695	8.8282828	0	0	0
3894	0.674669202	8.8282828	0	0	0
3895	0.681923710	8.8282828	0	0	0
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3897	0.696432725	8.8282828	0	0	0
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4199	0.710941740	9.5252525	0	0	0
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5636	0.253907764	13.0101010	0	0	0
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9896	0.689178217	22.7676768	0	0	0
9897	0.696432725	22.7676768	0	0	0
9898	0.703687232	22.7676768	0	0	0
9899	0.710941740	22.7676768	0	0	0
9900	0.718196248	22.7676768	0	0	0

9901	0.000000000	23.0000000	0	0	0
9902	0.007254508	23.0000000	0	0	0
9903	0.014509015	23.0000000	0	0	0
9904	0.021763523	23.0000000	0	0	0
9905	0.029018030	23.0000000	0	0	0
9906	0.036272538	23.0000000	0	0	0
9907	0.043527045	23.0000000	0	0	0
9908	0.050781553	23.0000000	0	0	0
9909	0.058036060	23.0000000	0	0	0
9910	0.065290568	23.0000000	0	0	0
9911	0.072545076	23.0000000	0	0	0
9912	0.079799583	23.0000000	0	0	0
9913	0.087054091	23.0000000	0	0	0
9914	0.094308598	23.0000000	0	0	0
9915	0.101563106	23.0000000	0	0	0
9916	0.108817613	23.0000000	0	0	0
9917	0.116072121	23.0000000	0	0	0
9918	0.123326628	23.0000000	0	0	0
9919	0.130581136	23.0000000	0	0	0
9920	0.137835643	23.0000000	0	0	0
9921	0.145090151	23.0000000	0	0	0
9922	0.152344659	23.0000000	0	0	0
9923	0.159599166	23.0000000	0	0	0
9924	0.166853674	23.0000000	0	0	0
9925	0.174108181	23.0000000	0	0	0
9926	0.181362689	23.0000000	0	0	0
9927	0.188617196	23.0000000	0	0	0
9928	0.195871704	23.0000000	0	0	0
9929	0.203126211	23.0000000	0	0	0
9930	0.210380719	23.0000000	0	0	0
9931	0.217635227	23.0000000	0	0	0
9932	0.224889734	23.0000000	0	0	0
9933	0.232144242	23.0000000	0	0	0
9934	0.239398749	23.0000000	0	0	0
9935	0.246653257	23.0000000	0	0	0
9936	0.253907764	23.0000000	0	0	0
9937	0.261162272	23.0000000	0	0	0
9938	0.268416779	23.0000000	0	0	0
9939	0.275671287	23.0000000	0	0	0
9940	0.282925794	23.0000000	0	0	0
9941	0.290180302	23.0000000	0	0	0
9942	0.297434810	23.0000000	0	0	0
9943	0.304689317	23.0000000	0	0	0
9944	0.311943825	23.0000000	0	0	0
9945	0.319198332	23.0000000	0	0	0

9946	0.326452840	23.0000000	0	0	0
9947	0.333707347	23.0000000	0	0	0
9948	0.340961855	23.0000000	0	0	0
9949	0.348216362	23.0000000	0	0	0
9950	0.355470870	23.0000000	0	0	0
9951	0.362725378	23.0000000	0	0	0
9952	0.369979885	23.0000000	0	0	0
9953	0.377234393	23.0000000	0	0	0
9954	0.384488900	23.0000000	0	0	0
9955	0.391743408	23.0000000	0	0	0
9956	0.398997915	23.0000000	0	0	0
9957	0.406252423	23.0000000	0	0	0
9958	0.413506930	23.0000000	0	0	0
9959	0.420761438	23.0000000	0	0	0
9960	0.428015945	23.0000000	0	0	0
9961	0.435270453	23.0000000	0	0	0
9962	0.442524961	23.0000000	0	0	0
9963	0.449779468	23.0000000	0	0	0
9964	0.457033976	23.0000000	0	0	0
9965	0.464288483	23.0000000	0	0	0
9966	0.471542991	23.0000000	0	0	0
9967	0.478797498	23.0000000	0	0	0
9968	0.486052006	23.0000000	0	0	0
9969	0.493306513	23.0000000	0	0	0
9970	0.500561021	23.0000000	0	0	0
9971	0.507815529	23.0000000	0	0	0
9972	0.515070036	23.0000000	0	0	0
9973	0.522324544	23.0000000	0	0	0
9974	0.529579051	23.0000000	0	0	0
9975	0.536833559	23.0000000	0	0	0
9976	0.544088066	23.0000000	0	0	0
9977	0.551342574	23.0000000	0	0	0
9978	0.558597081	23.0000000	0	0	0
9979	0.565851589	23.0000000	0	0	0
9980	0.573106097	23.0000000	0	0	0
9981	0.580360604	23.0000000	0	0	0
9982	0.587615112	23.0000000	0	0	0
9983	0.594869619	23.0000000	0	0	0
9984	0.602124127	23.0000000	0	0	0
9985	0.609378634	23.0000000	0	0	0
9986	0.616633142	23.0000000	0	0	0
9987	0.623887649	23.0000000	0	0	0
9988	0.631142157	23.0000000	0	0	0
9989	0.638396664	23.0000000	0	0	0
9990	0.645651172	23.0000000	0	0	0

9991	0.652905680	23.0000000	0	0	0
9992	0.660160187	23.0000000	0	0	0
9993	0.667414695	23.0000000	0	0	0
9994	0.674669202	23.0000000	0	0	0
9995	0.681923710	23.0000000	0	0	0
9996	0.689178217	23.0000000	0	0	0
9997	0.696432725	23.0000000	0	0	0
9998	0.703687232	23.0000000	0	0	0
9999	0.710941740	23.0000000	0	0	0
10000	0.718196248	23.0000000	0	0	0

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam_temporal,
                      newdata = ndvi_hour_values,
                      type = "link", # log-scale
                      se.fit = TRUE)

# Extract log RSS and standard errors
gam_log_rss <- predictions$fit
gam_se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
gam_log_rss_lower <- gam_log_rss - 1.96 * gam_se_log_rss
gam_log_rss_upper <- gam_log_rss + 1.96 * gam_se_log_rss

# Transform to RSS scale
rss <- exp(gam_log_rss)
rss_lower <- exp(gam_log_rss_lower)
rss_upper <- exp(gam_log_rss_upper)

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  gam_log_rss = gam_log_rss,
  gam_rss = exp(gam_log_rss),
  gam_log_rss_lower = as.numeric(gam_log_rss_lower),
  gam_log_rss_upper = as.numeric(gam_log_rss_upper)
)

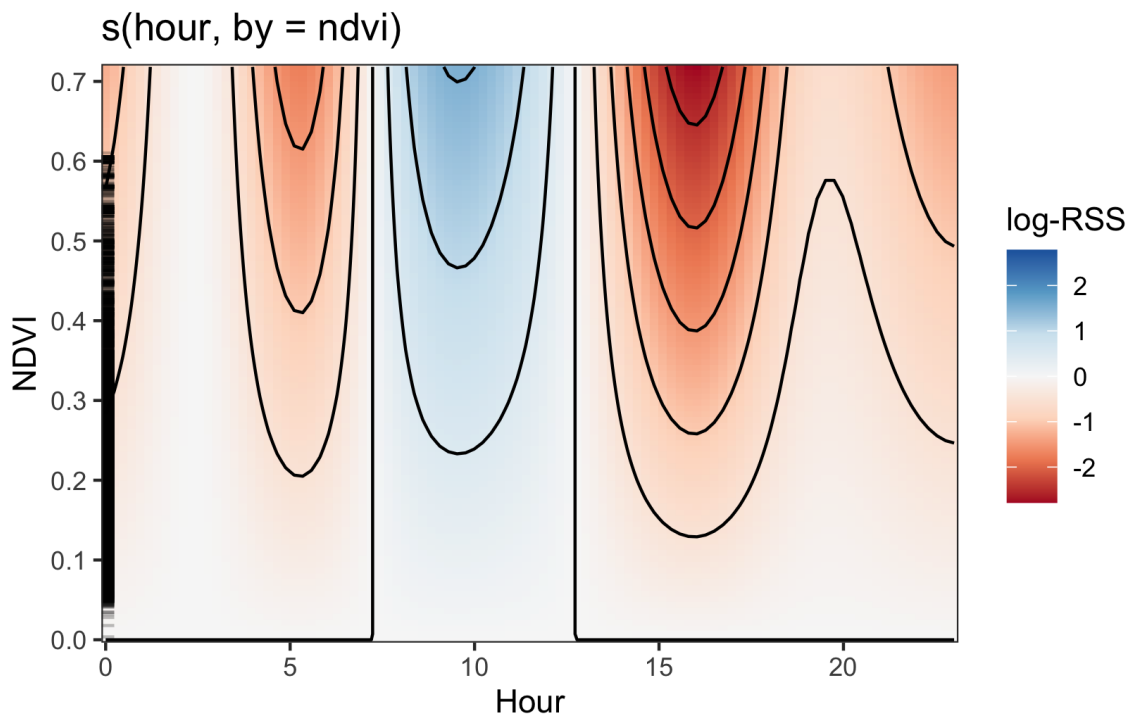
max_abs_colour <- max(abs(range(ndvi_hour_values$gam_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
             aes(x = hour, y = ndvi, fill = gam_log_rss)) +
  labs(title = "s(hour, by = ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
```

```

aes(x = hour, y = ndvi, z = gam_log_rss), colour = "black") +
scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
                      limits = c(-max_abs_colour, max_abs_colour)) +
coord_cartesian(expand = FALSE) +
geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
          aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
theme_bw()

```



```

ggsave(paste0(plot_save_path, "/ndvi_linear_gam_response_id", which_buffalo, ".png"),
        width = 150, height = 80, units = "mm", dpi = 600)

```

Fit 2D GAM smooth

```

# fit an ssf gam model
# As above, set the cyclic knots for the hour margin explicitly at 0 and 24
ssf_gam2d_temporal2 <- mgcv::gam(cbind(times, step_id_) ~
                                te(hour, ndvi, bs = c("cc", "cr")) + #, k = c(5, 5)
                                sl_ + log_sl + cos_ta,
                                knots = list(hour = c(0, 24)),
                                data = buffalo_ssf_data,

```

```

        family = cox.ph,
        weight = case_)

summary(ssf_gam2d_temporal2)

```

Family: Cox PH

Link function: identity

Formula:

```

cbind(times, step_id_) ~ te(hour, ndvi, bs = c("cc", "cr")) +
    sl_ + log_sl + cos_ta

```

Parametric coefficients:

	Estimate	Std. Error	z value	Pr(> z)
sl_	-2.009e-05	3.194e-05	-0.629	0.529
log_sl	6.985e-03	5.756e-03	1.214	0.225
cos_ta	-3.941e-03	1.505e-02	-0.262	0.793

Approximate significance of smooth terms:

	edf	Ref.df	Chi.sq	p-value
te(hour,ndvi)	12.25	13.83	346.1	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Deviance explained = 0.0413%

-REML = 23103 Scale est. = 1 n = 106678

```

AIC(ssf_gam2d_temporal2)

```

```

[1] 46156.49

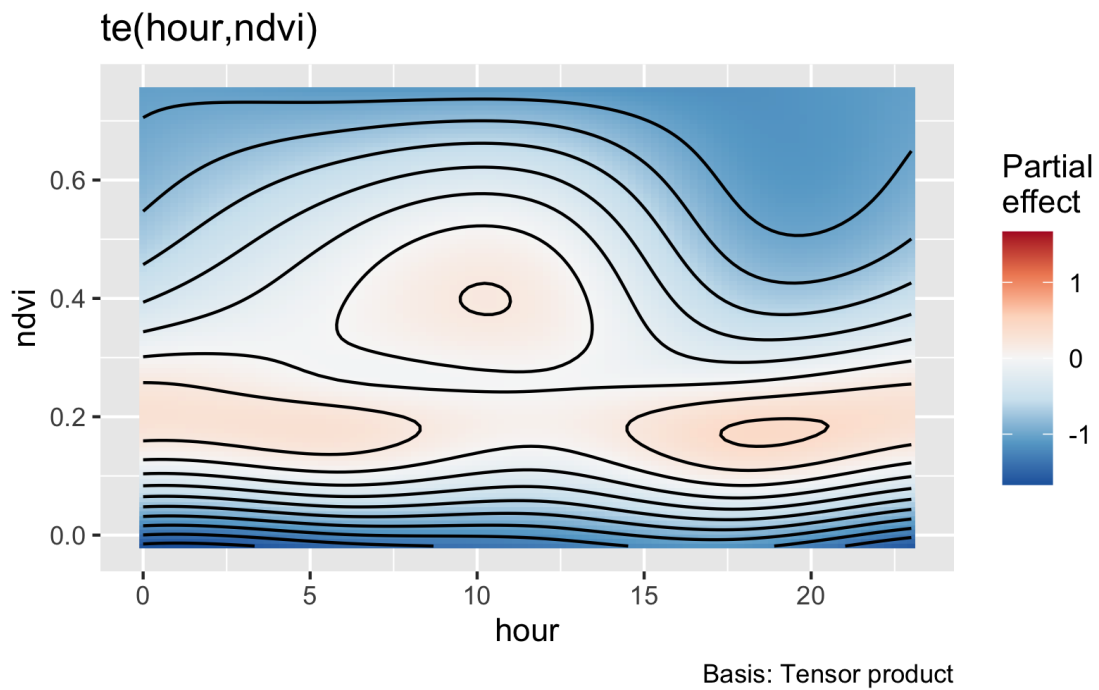
```

Plot the smooth selection surface

```

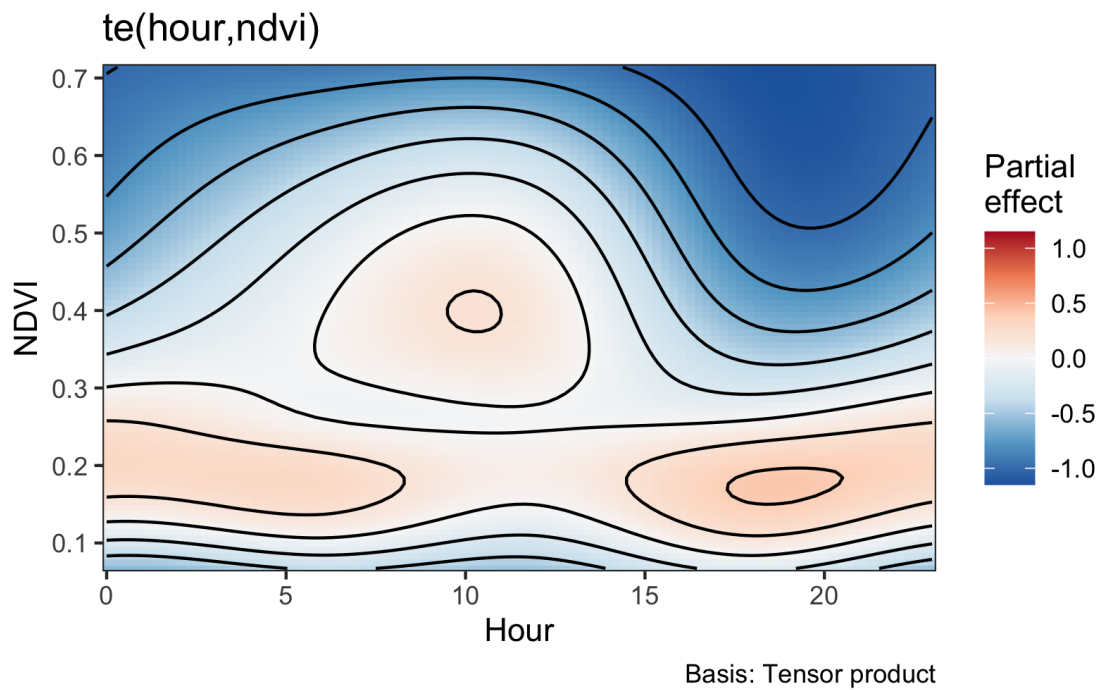
gratia::draw(ssf_gam2d_temporal2, rug = FALSE)

```



```
gam_smooths <- smooth_estimates(ssf_gam2d_temporal2)
filtered_gam_smooths <- gam_smooths %>% filter(ndvi > ndvi_quantiles[1] & ndvi < ndvi_quantiles[2])

gratia::draw(filtered_gam_smooths) +
  coord_cartesian(expand = FALSE) +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "Hour", y = "NDVI") +
  theme_bw()
```



```
ndvi_hour_values <- expand.grid(ndvi = ndvi_values,
                                hour = seq(0, 23, length.out = length(ndvi_values)))

ndvi_hour_values <- ndvi_hour_values %>% mutate(
  sl_ = 0,
  log_sl = 0,
  cos_ta = 0
)

predictions <- predict(ssf_gam2d_temporal2,
                        newdata = ndvi_hour_values,
                        type = "link", # log-scale
                        se.fit = TRUE)

# Extract log RSS and standard errors
gam_log_rss <- predictions$fit
gam_se_log_rss <- predictions$se.fit

# Calculate confidence intervals on log-scale
gam_log_rss_lower <- gam_log_rss - 1.96 * gam_se_log_rss
gam_log_rss_upper <- gam_log_rss + 1.96 * gam_se_log_rss

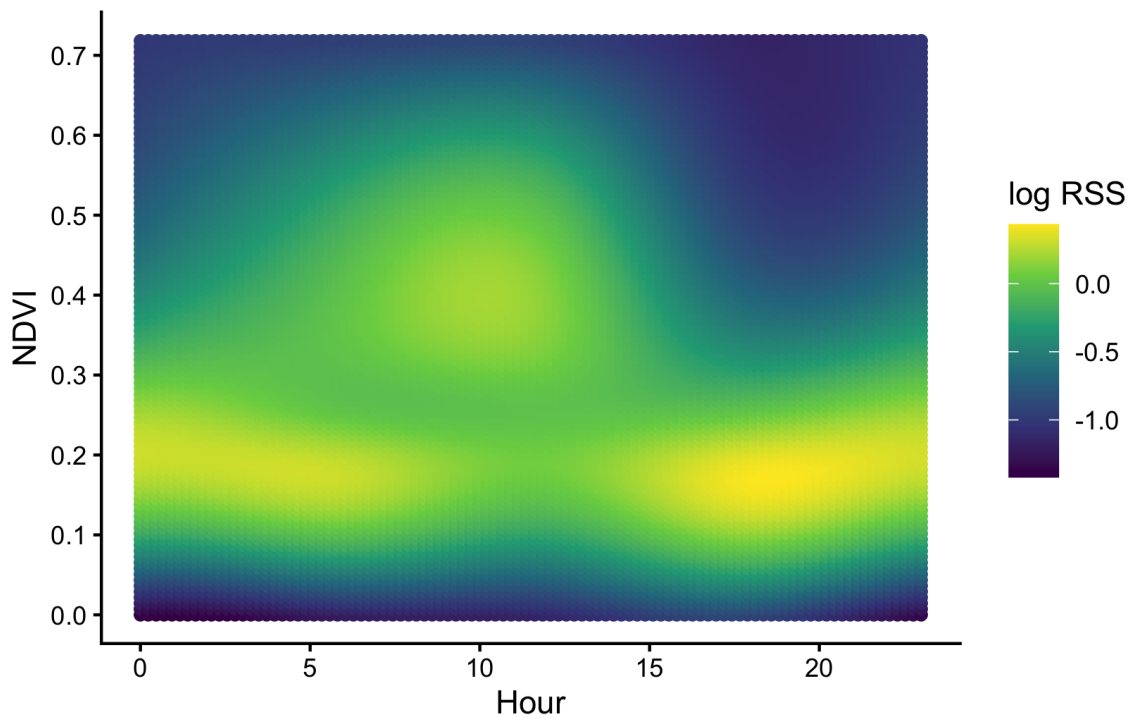
ndvi_hour_values <- ndvi_hour_values %>% mutate(
```

```

gam_log_rss = gam_log_rss,
gam_rss = exp(gam_log_rss),
gam_log_rss_lower = as.numeric(gam_log_rss_lower),
gam_log_rss_upper = as.numeric(gam_log_rss_upper)
)

# plot the selection surface
ggplot(data = ndvi_hour_values) +
  geom_point(aes(x = hour, y = ndvi, colour = gam_log_rss)) +
  labs(x = "Hour", y = "NDVI", colour = "log RSS") +
  scale_colour_viridis_c() +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_classic()

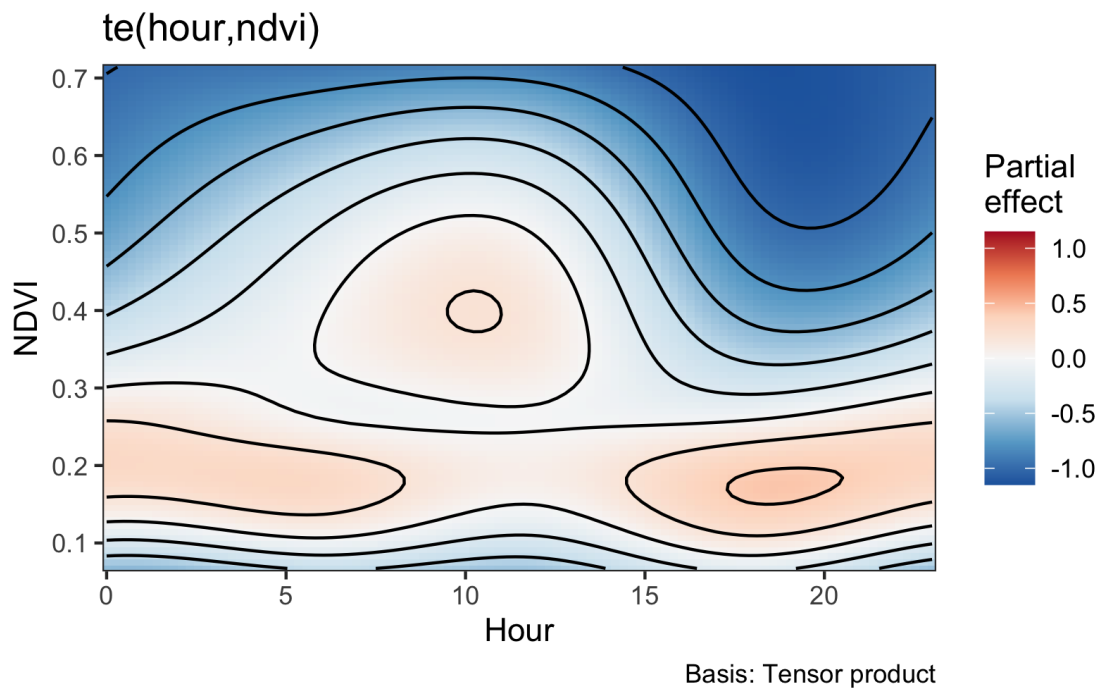
```



```

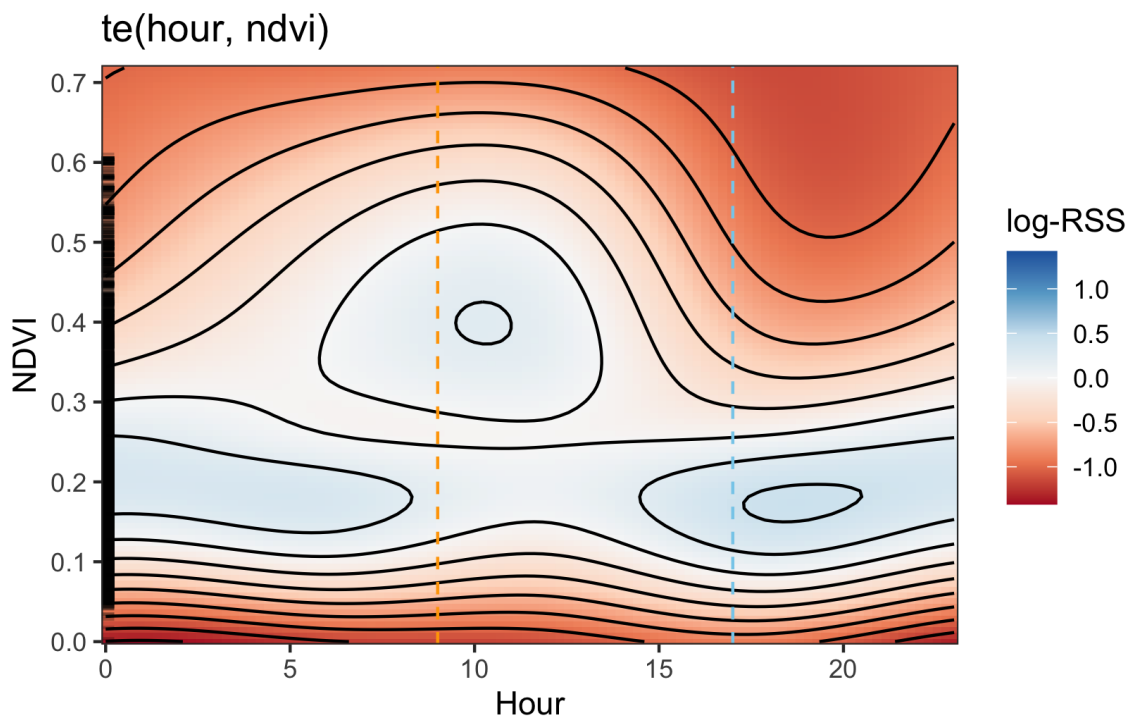
gratia::draw(filtered_gam_smooths) +
  coord_cartesian(expand = FALSE) +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  labs(x = "Hour", y = "NDVI") +
  theme_bw()

```



```
max_abs_colour <- max(abs(range(ndvi_hour_values$gam_log_rss)))

# plot the selection surface
ggplot() +
  geom_raster(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, fill = gam_log_rss)) +
  labs(title = "te(hour, ndvi)", x = "Hour", y = "NDVI", fill = "log-RSS") +
  geom_contour(data = ndvi_hour_values,
    aes(x = hour, y = ndvi, z = gam_log_rss), colour = "black") +
  geom_vline(xintercept = 9, linetype = "dashed", colour = "orange") +
  geom_vline(xintercept = 17, linetype = "dashed", colour = "skyblue") +
  scale_fill_distiller(palette = "RdBu", type = "div", direction = 1,
    limits = c(-max_abs_colour, max_abs_colour)) +
  coord_cartesian(expand = FALSE) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
    aes(y = ndvi), alpha = 0.25, length = unit(0.015, "npc"), sides="l") +
  scale_y_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  theme_bw()
```



```
ggsave(paste0(plot_save_path, "/ndvi_tensor_gam_response_id", which_buffalo, ".png"),
       width = 150, height = 80, units = "mm", dpi = 600)
```

Pull out a response curve from one of the hours

```
# Get predictions on log-scale with standard errors
predictions <- predict(ssf_gam2d_temporal2,
                      newdata = pred_data_hour9_17,
                      type = "link", # log-scale
                      se.fit = TRUE)

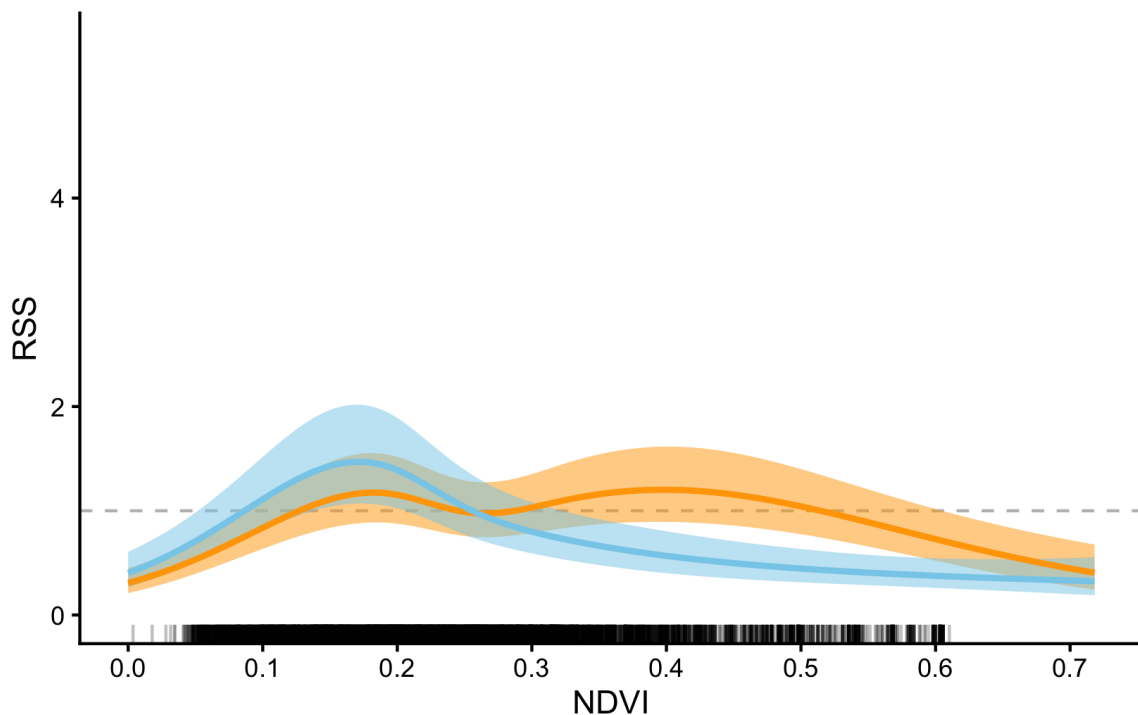
pred_data_hour9_17 <- pred_data_hour9_17 %>%
  mutate(
    # Extract log RSS and standard errors
    log_rss = predictions$fit,
    se_log_rss = predictions$se.fit,
    # Calculate confidence intervals on log-scale
    log_rss_lower = log_rss - 1.96 * se_log_rss,
    log_rss_upper = log_rss + 1.96 * se_log_rss,
    # Transform to RSS scale
    rss = exp(log_rss),
```

```

    rss_lower = exp(log_rss_lower),
    rss_upper = exp(log_rss_upper)
  )

# plot the response curve
ggplot(data = pred_data_hour9_17) +
  geom_hline(yintercept = 1, linetype = "dashed", colour = "grey") +
  geom_ribbon(aes(x = ndvi,
                 ymin = rss_lower, ymax = rss_upper,
                 fill = as.factor(hour)),
            alpha = 0.5) +
  geom_line(aes(x = ndvi, y = rss, colour = as.factor(hour)),
            linewidth = 1) +
  geom_rug(data = buffalo_ssf_data %>% filter(case_ == T),
           aes(x = ndvi), alpha = 0.25, sides="b") +
  scale_colour_manual(values = c("orange", "skyblue")) +
  scale_fill_manual(values = c("orange", "skyblue")) +
  scale_x_continuous(breaks = seq(-0.2, 0.8, by = 0.1)) +
  scale_y_continuous(limits = c(0, 5.5)) +
  labs(x = "NDVI", y = "RSS") +
  theme_classic() +
  theme(legend.position = "none")

```



```
ggsave(paste0(plot_save_path, "/ndvi_gam2d_response_sample_hour_id", which_buffalo, ".png"),  
       width = 70, height = 80, units = "mm", dpi = 600)
```

References

Klappstein, Natasha J, Théo Michelot, John Fieberg, Eric J Pedersen, and Joanna Mills Flemming. 2024. "Step selection functions with non-linear and random effects." *Methods in Ecology and Evolution*, ahead of print, June 24. <https://doi.org/10.1111/2041-210x.14367>.